



Laser welding of aluminum/steel dissimilar metals and control of intermetallic compounds: a review

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Abstract

In the context of automotive lightweighting, aluminum (Al)/steel dissimilar welding represents an effective approach toward achieving this goal. The primary challenge in welding these materials lies in the formation of brittle intermetallic compounds (IMCs) due to the significant differences in their physical and chemical properties, which directly impacts the mechanical performance of the welded joints. This review examines the mechanisms of IMCs formation during laser welding of Al/steel dissimilar metals. It also presents the latest research advancements in controlling IMCs through thermal input and the addition of filler materials, in laser keyhole welding and laser welding-brazing. Furthermore, this paper also introduces the application of other advanced laser-related welding techniques, such as laser-arc hybrid welding (LAHW) and laser-assisted friction stir welding (LAFSW), in joining Al and steel is discussed. Additionally, the utilization of numerical simulation technologies in the study of Al/steel laser welding is summarized. Finally, the future research directions in this field are outlined.

Keywords Aluminium · Steel · Laser welding · Mechanical property · Intermetallic compounds (IMCs)

1 Introduction

With the intensifying issues of energy consumption and environmental pollution, achieving energy conservation and emission reductions has become an urgent matter in the automotive manufacturing industry [1]. One effective approach to this end is the lightweighting of automobiles. Studies show that for every 10% reduction in vehicle weight, fuel consumption can decrease by 6–8%, and carbon dioxide emissions can be cut by 13%. Similarly, for electric vehicles, every 10% reduction in weight can increase their driving range by 5–6% [2, 3]. Commonly used lightweight materials include aluminum alloys, titanium alloys, magnesium alloys, carbon fiber, and engineering plastics. Among these, aluminum alloys stand out due to their high specific strength, good ductility, and corrosion resistance, making them the most widely used lightweight material in automotive

manufacturing [4]. Replacing traditional high-strength steel bodies with Al/steel composite bodies can achieve the goal of automotive lightweight design while maintaining the necessary structural strength.

Among these, the welding of Al/steel dissimilar metals has become a key aspect in the application of aluminum alloys in lightweight vehicle bodies. Due to the significant differences in physicochemical properties and the low solid solubility between Al and Fe, there is a tendency to form large quantities of brittle IMCs during the welding process. This severely impacts the mechanical properties of the joint, thereby limiting the application of Al/steel dissimilar welded components in industry [5]. Consequently, suppressing the formation of IMCs at the interface and regulating their distribution on the Al/Fe interface to enhance the mechanical properties of the joint has become an urgent challenge that needs to be addressed to achieve high-quality and efficient welding of Al/steel dissimilar metals.

Extensive research has been conducted by scholars both domestically and internationally on the welding of Al/steel dissimilar metals. Zhang et al. [6] utilized a medium-frequency DC resistance spot welding technique, employing a 4047 AlSi12 foil as an interlayer to join galvanized high-strength steel (H220YD) with Al alloy (6008-T66). They investigated the effect of interlayer thickness on the

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microstructure and mechanical properties of the weld joint. The results indicated that an IMCs layer composed of $\text{Fe}_2(\text{Al},\text{Si})_5$ and $\text{Fe}_4(\text{Al},\text{Si})_{13}$ formed in the interface region of the weld joint, with the thickness and morphology of this layer varying according to the thickness of the interlayer. Yazdipour et al. [7] used friction stir welding to achieve the joining of Al alloy (5083-H321) and stainless steel (316L), examining the impact of process parameters on the microstructure and mechanical properties of the joint. The results demonstrated that defects such as cracks and pores at the interface are the primary cause of decreased shear strength in Al/steel lap joints. Yang et al. [8] used explosive welding technology to obtain a mesh-type bonding interface by pre-fabricating a dovetail groove on the substrate, and the mesh interface achieved a pore-free metallurgical bonding. The mechanical properties of the formed welded joints are better than those of conventional Al/steel explosive welded joints. Although the above studies have improved the mechanical properties of welded joints to varying degrees, there are still a variety of Fe–Al brittle phases in the weld zone, which seriously affects the mechanical properties of welded joints. In addition, processes such as friction stir welding and explosive welding have strict requirements on the geometry of the weldment, which have great limitations in industrial production applications.

Laser welding is an efficient and precise welding method utilizing a high-energy-density laser beam as a heat source. Due to its advantages of concentrated energy, low thermal input, and rapid welding speeds, the metallurgical reactions during the welding process are confined to a very short duration. This not only effectively inhibits the formation of brittle IMCs but also alleviates the segregation issues in heterogeneous joints, thus achieving high-quality welds [9–11]. Currently, the use of laser welding of Al/steel dissimilar metals has become a focus of research within the industry. Among these studies, the control of IMCs remains one of the core issues in Al/steel laser welding. This paper reviews the formation mechanisms of IMCs during the laser welding of Al/steel dissimilar metals and summarizes the research on IMCs control. Based on the control of IMCs, it introduces new technologies for the laser welding of dissimilar Al/steel, as well as the application of numerical simulation technologies in this field.

2 Joint metallurgical connection

2.1 Laser weldability of Al/steel

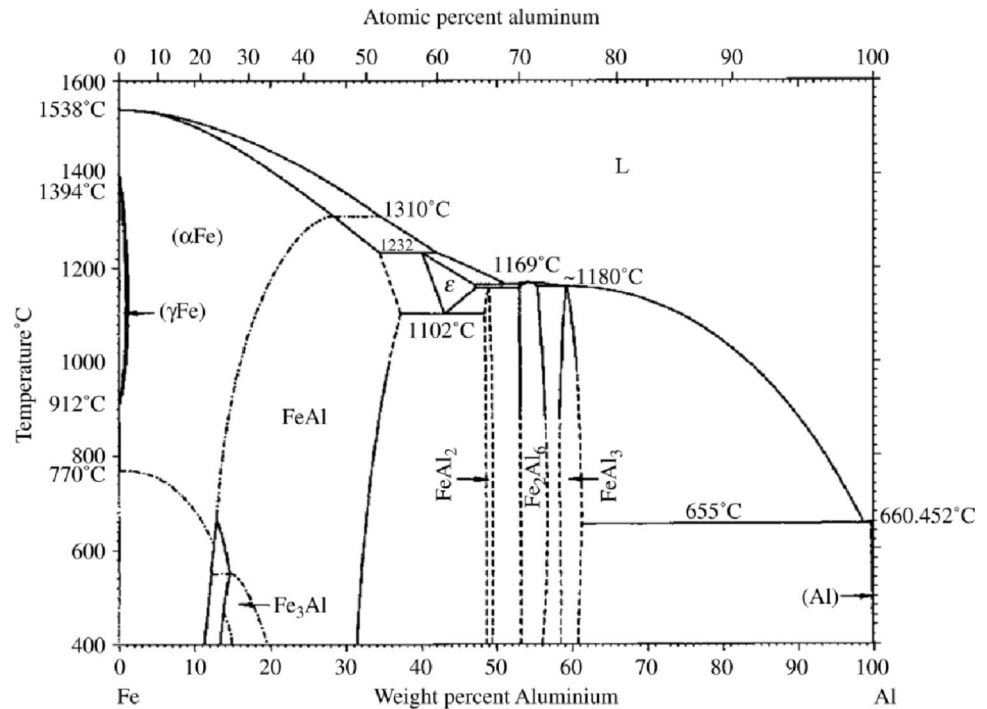
The laser weldability of Al/steel dissimilar metals mainly depends on the differences in physical and chemical properties. The greater the difference, the more difficult the welding of Al/steel dissimilar metals [12]. Table 1 [13]

Table 1 Comparison of physical properties of Fe and Al [13]

Properties	Fe	Al
Density/ $\text{g}\cdot\text{cm}^{-3}$	7.86	2.69
Melting point/ $^{\circ}\text{C}$	1538	660
Boiling point/ $^{\circ}\text{C}$	2750	2327
Thermal conductivity/ $\text{W}\cdot\text{m}^{-1}\cdot\text{K}^{-1}$	80	237
Thermal expansion coefficient/ $10^{-6}\cdot\text{K}^{-1}$	11	24
Surface tension/ $\text{N}\cdot\text{m}^{-1}$	1.872	0.914
Elastic modulus/GPa	210	70
Specific heat/ $\text{J/Kg}\cdot\text{K}$	460	900

lists the physical properties of pure iron (Fe) and pure aluminum (Al). From Table 1, it can be observed that the melting point of Fe is 1538°C , whereas Al has a melting point 878°C lower than that of iron. During the laser welding process, the lower melting point aluminum melts while the steel often remains in a solid phase heating state. Due to the significant difference in density between aluminum and steel, when both are in a liquid state, the aluminum liquid suspends on the surface of the steel, leading to non-uniform composition in the cooled and solidified weld seam, thereby reducing the mechanical properties of the welded joint. Additionally, the differences between aluminum and steel in terms of linear expansion coefficients, thermal conductivity, and specific heat capacities can cause uneven temperature distribution across different areas during the welding process. After welding, the joint may experience varying degrees of deformation and contain significant residual stresses, making it easier for cracks to form at the interface and propagate outward.

According to the Fe–Al binary phase diagram [14] (Fig. 1), at high temperatures, aluminum and iron can form solid solutions, eutectic structures and IMCs. However, during cooling, the solid solubility of iron in solid aluminum is extremely low, with a solubility of merely 0.01–0.022% when the temperature ranges from 225°C to 600°C [15]. Aluminum alloys containing trace amounts of Fe can form various brittle intermetallic compounds (IMCs), such as FeAl , Fe_3Al , FeAl_7 , FeAl_2 , Fe_2Al_5 , and FeAl_{13} , among others. Among these, Fe_2Al_5 exhibits the highest brittleness, which can lead to reduced toughness and even cracking in Al/steel welded joints, severely impacting their mechanical properties. Additionally, this IMC features a significant number of vacancies along its c-axis, providing a pathway for rapid diffusion of Al atoms, which results in the Fe_2Al_5 layer being thicker than other IMC layers [16]. Similarly, research by Wan et al. [17] indicates that the reason why the Fe_2Al_5 layer is thicker than the FeAl_3 layer is due to the presence of crystal defects, which accelerate its growth. This is because the IMC possesses $\approx 30\%$ vacancies along its c-axis, providing a rapid diffusion path and facilitating anisotropic growth.

Fig. 1 Fe-Al binary phase diagram [14]

Therefore, to successfully achieve a high-quality weld joint between Al and Fe, ensuring a reliable connection between the two, it is not only necessary to select appropriate base materials, welding methods, and optimal welding process parameters, but also to address the core issue of the formation and control of Fe-Al IMCs.

2.2 IMCs formation mechanism

During laser welding of aluminum to steel, some intermetallic compounds (IMCs) form at the Al/steel interface, enabling an effective connection between the two materials; however, excessive IMCs can severely deteriorate the welded joints. These IMCs primarily result from the diffusion of Al and Fe atoms. At high temperatures, Al and Fe atoms migrate, forming a diffusion layer. Within this diffusion layer, Al and Fe atoms interchange positions, thereby altering the original composition of the materials. As diffusion progresses, aluminum and steel atoms rearrange at the interface, undergoing chemical reactions to form IMCs [18], as illustrated in Fig. 2.

The formation and growth of Al/Fe IMCs mainly depend on thermodynamics and diffusion kinetics. The driving force of chemical reaction can be expressed by the change of Gibbs free energy (ΔG) (Eq. (1)) [19]:

$$\Delta G = \Delta H - (T \cdot \Delta S) \quad (1)$$

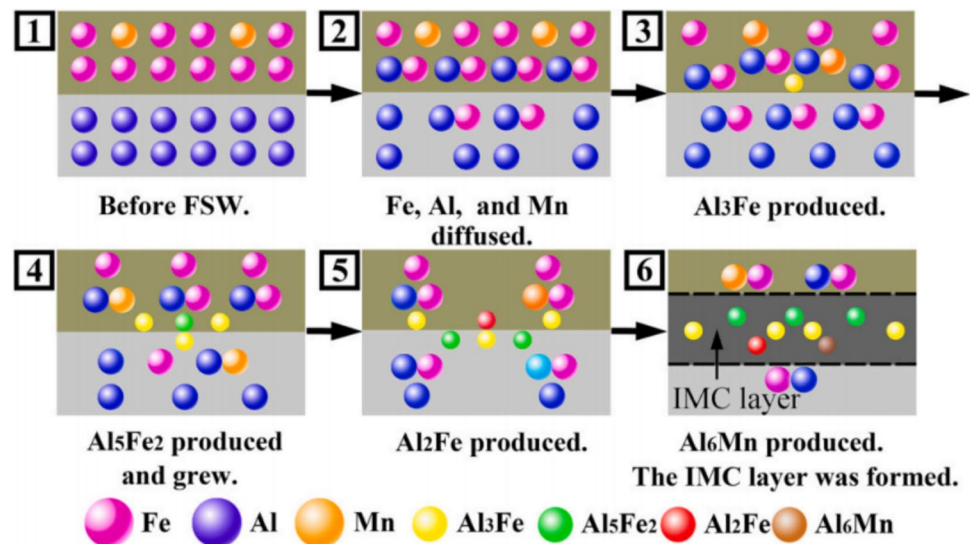
where ΔH ($\text{J} \cdot \text{mol}^{-1}$) is enthalpy change, T (K) is temperature, and ΔS ($\text{J} \cdot (\text{K} \cdot \text{mol})^{-1}$) is entropy change. The enthalpy of formation ($\Delta_f H(T)$) and entropy ($\Delta_f S(T)$) at a certain temperature are calculated from (Eq. (2)) and (Eq. (3)) [20]:

$$\Delta_f H(T) - \Delta_f H(298\text{K}) = \int_{298}^T \Delta C_p dT \quad (2)$$

$$\Delta_f S(T) - \Delta_f S(298\text{K}) = \int_{298}^T \Delta C_p \frac{dT}{T} \quad (3)$$

where $\Delta_f H(298\text{K})$ is the standard enthalpy of formation at 298 K ($\text{J} \cdot \text{mol}^{-1}$), $\Delta_f S(298\text{K})$ is the standard entropy at 298 K ($\text{J} \cdot (\text{K} \cdot \text{mol})^{-1}$), and ΔC_p is the difference in specific heats under constant pressure between reactants and products ($\text{J} \cdot (\text{K} \cdot \text{mol})^{-1}$). The crystal structures and thermodynamic constants of various Fe/Al IMCs are presented in Table 2 [21, 22]. Based on the above, chemical reactions tend to form phases with lower Gibbs free energy (ΔG); thus, the sequence of IMCs formation during non-equilibrium reactions can be predicted, which is FeAl_3 , Fe_2Al_3 , FeAl_2 , FeAl and Fe_3Al [23].

Furthermore, the growth of IMCs is controlled by diffusion mechanisms; for example, the relationship between the thickness of Fe_2Al_3 and reaction time follows a parabolic rate law [25–29], where the thickness of the reaction layer is

Fig. 2 The evolution mechanism of Fe/Al IMCs [18]**Table 2** Crystal structure and thermodynamic constants of different types of Fe/Al IMCs [19, 20, 24]

IMCs types	Crystal structure	$\Delta H_{298k}(\text{J}\cdot\text{mol}^{-1})$	$\Delta S_{298k}(\text{J}\cdot(\text{K}\cdot\text{mol})^{-1})$	$\Delta G_{298k}(\text{J}\cdot\text{mol}^{-1})$
Fe_3Al	Cubic crystal system	−67,660	−90	−40,840
FeAl	Cubic crystal system	−58,950	−16	−54,182
FeAl_2	Triclinic crystal system	−102,577	−33.2	−92,683.4
Fe_2Al_5	Rhombic crystal system	−253,971	−85.9	−228,372
FeAl_3	Monoclinic crystal system	−142,770	−50.6	−127,691.2
$\text{Fe}_4\text{Al}_{13}$	Monoclinic crystal system	−603,814	−75.84	−581,215.3

plotted as a function of the square root of the dipping time. The growth rate constant K of IMCs relates to temperature T according to the Arrhenius equation [28]:

$$K = K_0 \exp\left(-\frac{Q}{RT}\right) \quad (4)$$

where K_0 is a constant, Q is the activation energy (J) for the growth of the IMCs layer, and R is the gas constant ($8.314 \text{ J}\cdot\text{K}^{-1}\cdot\text{mol}^{-1}$). Typically, the value of Q is derived by fitting experimental data obtained at different temperatures.

Based on the diffusion-controlled nature of IMCs layer growth, the equation of layer thickness x as a function of time t can be used [30]:

$$x = k't^n \quad (5)$$

where x is the thickness of the IMCs layer, k' is the growth rate constant, n is the growth index of the composite layer, and the growth index of the composite layer is expected to be 0.5 under normal diffusion control growth.

Pan [31] analyzed the formation process of IMCs layer by finite element prediction and thermodynamic calculation, combined with the diffusion coefficient of different phase atoms. Fe_2Al_5 exhibits a faster diffusion rate,

leading to its formation in regions of high, medium and low welding energies. Due to the lower content of Fe atoms in $\text{Fe}_4\text{Al}_{13}$, it primarily forms under different temperature conditions, Fig. 3 illustrates the EBSD mapping images of IMCs at the interface center.

The formation of IMCs is primarily driven by the thermal diffusion of Al and Fe atoms and interfacial chemical reactions. Key factors influencing IMC formation include welding heat input, cooling rate, the physical and chemical properties of materials, and the effects of alloying elements. The presence of IMCs significantly impairs the mechanical properties of the joint. For instance, Fe_2Al_5 exhibits high brittleness and hardness, making it prone to causing interfacial cracks and reducing joint toughness, thus significantly increasing the risk of fracture under tensile or shear loads. An excessively thick IMCs layer can also cause stress concentration, weakening the joint's fatigue resistance. Reducing IMCs formation is the primary goal for achieving high-quality Al/Fe welding. Therefore, a thorough understanding of IMCs formation mechanisms and the development of efficient control strategies are essential for advancing the industrial application of Al/Fe dissimilar welding technology.

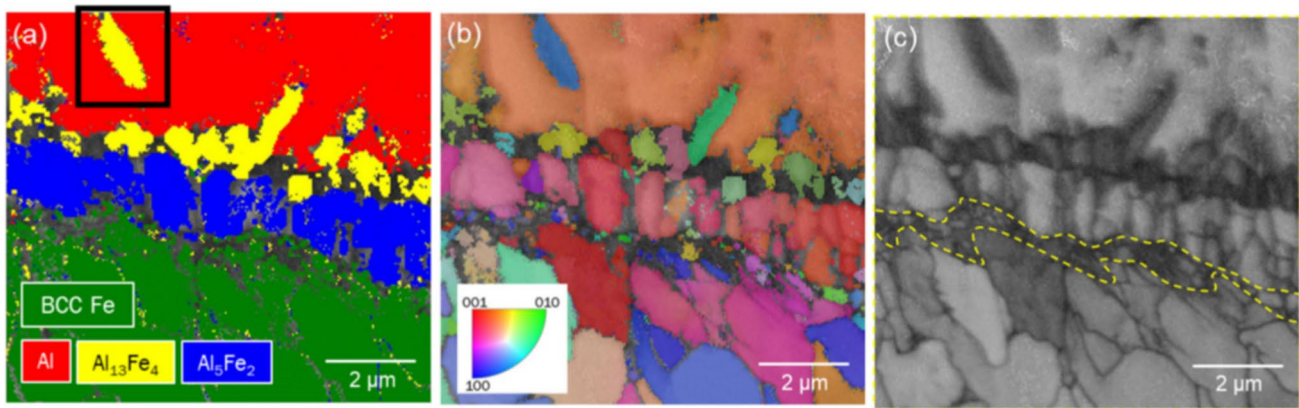


Fig. 3 EBSD mapping of IMCs at the interface center: (a) phase distribution; (b) OIMs; (c) band comparison [31]

3 IMCs regulation

How to suppress the formation and regulate the growth of IMCs is crucial for laser welding of dissimilar Al/steel metals, directly impacting the mechanical properties of the welded joints. The control of IMCs in laser welding of dissimilar Al/steel metals can be approached from both energetic and metallurgical perspectives [16]. Regarding the regulation of IMCs, the following sections detail the advancements in laser keyhole welding and laser welding-brazing concerning the welding heat input and the addition of filler materials.

3.1 Laser keyhole welding

During the laser welding process, depending on the mode of heat conduction, it can be classified into heat conduction welding and keyhole welding. The former has a lower power density, slower welding speed, and a smaller depth-to-width ratio; whereas the latter has a higher power density, faster welding speed, and a larger depth-to-width ratio [9]. In heat conduction welding, the laser irradiates and heats the surface of the workpiece, and the heat penetrates into the interior of the workpiece through the conduction mode, so that it melts to form a molten pool. In laser keyhole welding, it is achieved by forming a “keyhole” effect. At high energy density, the surface metal rapidly vaporizes to form metal vapor, and the recoil pressure brought by the metal vapor causes the surface of the liquid metal to collapse to form small holes. The laser beam continuously enters the hole, most of the energy is absorbed, a small part is reflected, and the outer wall of the hole is surrounded by flowing molten metal. As the laser beam moves, the liquid metal continuously fills the hole to realize welding.

When keyhole welding occurs, the energy coupling increases. Due to the high energy density, the welding speed can be increased and the production efficiency can be

improved. In addition, low heat input can reduce the transverse expansion of the heat affected zone and the fusion zone. In industry, laser welding is mainly concentrated on keyhole welding. However, many researchers have found that IMCs are still formed at the Fe/Al interface during laser keyhole welding of Al/steel dissimilar materials. Next, the research content of IMCs regulation in laser keyhole welding of Al/steel is introduced.

3.1.1 Control of heat input

Heat input is one of the most critical parameters affecting welding quality and is also one of the most controllable parameters. Rational adjustment of welding heat input can improve the chemical composition of the weld seam, the dilution rates of different base materials, and reduce the formation of joint defects [32]. During the laser welding of dissimilar metals such as Al/steel, appropriate heat input can reduce the diffusion between Fe and Al atoms, thereby decreasing the formation of IMCs.

Guan et al. [10] Al alloy using a fiber laser. The Fe-Al interfacial layer was analyzed using SEM–EDS, revealing that regardless of variations in the laser welding parameters, there would always be “islands” of solute with higher Al content, forming the IMCs layer. EDS analysis indicated that this region primarily consisted of Fe_2Al_5 and Fe_3Al phases. Additionally, With the decrease of heat input, the formation of brittle Fe-Al phase is inhibited and the mechanical strength of the joint is improved. Cui [33] optimized process parameters such as welding speed and laser beam offset. The principle of laser welding is illustrated in Fig. 4a,b. The results demonstrated that the welding speed and laser beam offset influenced the morphology of the weld and the distribution, morphology and thickness of Fe-Al IMCs. Figure 4c shows the EBSD phase mapping of the Al/Fe joint interface with a laser beam offset of 0.3 mm. The needle-like, jagged $\text{Fe}_4\text{Al}_{13}$ phase and lath-like Fe_2Al_5 phase are formed at the

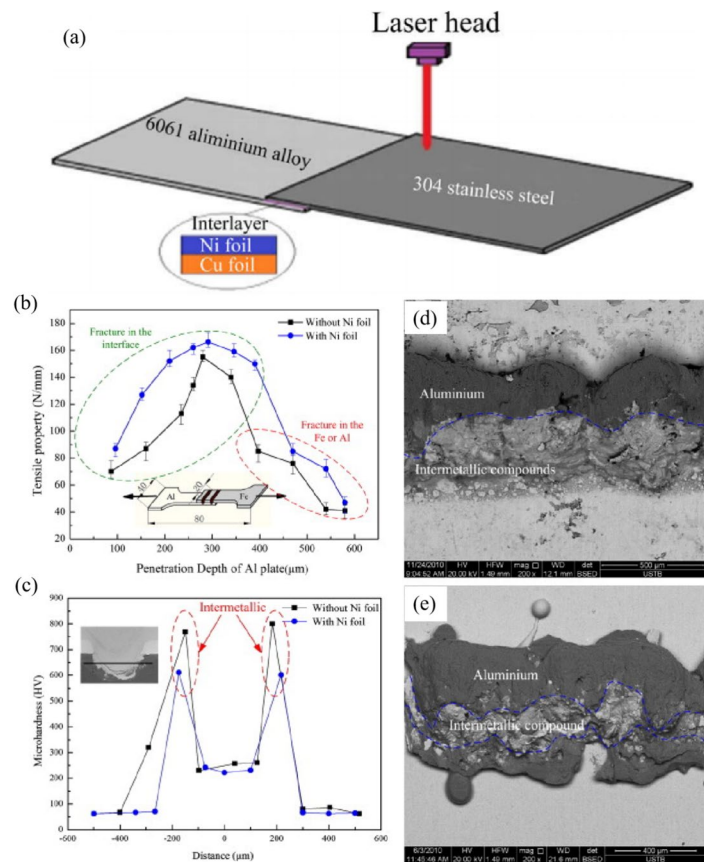
Table 3 Study on the effect of welding heat input on the composition and strength of IMCs in Al/steel joints

Welding mode	BM	Welding parameters	Joint forms	Filler metal	IMCs phase	Mechanical property	Ref
Laser keyhole welding	DC-04 steel (1 mm)/6016 Al (1 mm)	1.4 kW laser power, 40 mm/s weld speed, 0 mm defocusing value	Lap joint	Si,Zn and Ni powders 1:1:1	Fe ₂ Al ₅ , Fe ₃ Al	102 N/mm max tensile strength	[10]
	Q235 steel (6 mm)/5083 Al (6 mm)	P:3.25 kW f:0 mm V:1.0 m/min laser beam offset (Δd):0.6 mm	Butt joint	None	Fe ₂ Al ₅ , Fe ₃ Al and Fe ₄ Al ₁₃	107 MPa max tensile strength	[33]
	304 stainless steel (1.8 mm)/5083 Al (1.5 mm)	P:1.87 kW V:3.0 m/min f:0 mm	Lap joint	None	Fe ₂ Al ₅ , Fe ₄ Al ₁₃	137.1 N/mm max tensile strength	[34]
	St14 steel (0.8 mm)/5754 Al (2 mm)	P:1.43 kW pulse duration:5 ms V:4 mm/s	Lap joint	None	FeAl ₂ , Fe ₂ Al ₅ and FeAl ₃	/	[11]
Laser welding-brazing	Zinc-coated DC 04 steel (1.2 mm)/6016 Al (1 mm)	P: 3 kW V:1 m/min	Lap joint	None	Fe ₂ Al ₅	90 MPa max. shear strength	[35]
	JSC270CC steel (0.8 mm)/A6111-T4 Al (1.2 mm)	CW laser power:390W PW laser power:3.61kW V:0.06 m/min	Lap joint	None	Fe ₃ Al and FeAl	128 MPa max. shear strength	[36]
	DP590 steel (1.2 mm)/7075 Al (1 mm)	P:1.4kW V:0.7 m/min	Lap joint	None	Fe ₂ Al ₅ and FeAl ₃	123.7 MPa max. tensile strength	[37]

powders, coatings, etc.) to regulate the formation and distribution of IMCs at the Fe/Al interface. Zhou et al. [38] compared the microstructure and mechanical properties of the laser keyhole welded lap joints of DC56D + ZF steel and 6016 aluminum alloy with or without the addition of Mn, Zr, and Sn powders. The study found that the thickness of the IMCs layer from greatest to least was no powder, Mn, Zr, Sn, with the smallest IMCs layer thickness being approximately 10–13 μm; the average shear strength of the Al/Fe joints from least to greatest was no powder, Mn, Zr, Sn, with the highest average maximum shear strength reaching 62.17 MPa. In the presence of Sn powder, no significant pores, cracks, or other defects were observed. Additionally, a FeSn phase formed at the Fe/Al interface. The structure of FeSn is more stable than FeAl, with better ductility, thus significantly improving the mechanical properties of the welded joint. Yang [39] used 0.1 mm pure Cu foil as the interlayer material to carry out laser butt welding test on 5083 Al alloy and 08A1 steel. The results showed that after adding the Cu foil, the macroscopic morphology of the weld seam transformed from

T-shape to X-shape. As a transition layer between steel and aluminum, Cu foil has a gradual transition between Fe/Al interface, which reduces the formation of coarse lath brittle Fe₄Al₁₃ and improves the mechanical properties. The static tensile load of the weld is 2781N, which is 45% higher than that without Cu foil. Cao et al. [40] performed laser keyhole welding of 6061 Al alloy to 304 stainless steel by adding a composite Cu/Ni interlayer, placing Ni foil and Cu foil adjacent to the steel and Al respectively, as shown in Fig. 5a. The results indicated that with a 0.05 mm Cu foil and a 0.1 mm Ni foil serving as the interlayer, numerous new Cu-Al binary phases were observed near the aluminum alloy, effectively suppressing the formation of brittle Fe-Al binary phases. When the thickness of the copper foil was changed to 0.02 mm, the Fe-Al interface reaction zone exhibited a gear-like shape, and the maximum shear force of the welded joint reached 1754.72 N. Chen et al. [41] successfully joined 201 stainless steel to 5052 aluminum alloy by inserting a 0.1 mm thick pure Ni interlayer and employing laser keyhole welding. They comparatively investigated the

Fig. 5 (a) Schematic diagram of laser welding of aluminum steel; (b, c) tensile properties and microhardness of the joints; (d, e) the fracture morphologies of without Ni and with Ni foil [40, 41]



microstructure and mechanical properties of the welded joints with and without an interlayer. The authors noted that, compared to joints without an interlayer, the addition of the Ni foil changed the IMCs at the Fe–Al interface from Fe_2Al_5 and FeAl_3 phases to $\text{Al}_{0.9}\text{Ni}_{1.1}$ and FeAl_3 phases. Figure 5b–e compares the tensile properties, microhardness and fracture morphology of joints with or without the Ni interlayer. Due to the softening effect of $\text{Al}_{0.9}\text{Ni}_{1.1}$ and the reduction in the volume fraction of FeAl_3 , the corresponding microhardness decreased, while the strength of the joint improved.

In addition, H.R. Kotadia [42] carried out laser lap welding of IF steel and 1050 aluminum alloy with Zn coating, and studied the influence of welding parameters on the formation of IMCs and the effect on the joint. As shown in Fig. 6a,b, under keyhole welding mode, samples with higher power or power density exhibited the greatest depth of fusion, increasing the degree of mixing between Fe and Al, which led to excessive IMCs formation and cracking in these samples. Chen et al. [43] prepared Cu coatings of varying thicknesses on steel substrates by controlling the electroplating time for use in laser keyhole welding to Al. Their results showed that the joint strength increased with the thickness of the Cu coating due to the promotion of a

lesser amount of less brittle Fe-rich IMCs. Table 4 summarizes recent studies on the regulation of IMCs composition and strength in Al/Fe joints by adding filler materials.

3.2 Laser welding-brazing

Laser welding-brazing is based on the heat conduction mode of laser. During the welding process, the steel remains solid, only the low melting point aluminum alloy base metal, filler metal and flux are melted, so that the molten aluminum alloy and filler metal are wetted and spread on the surface of the steel base metal to form a brazed joint. Compared with laser keyhole welding, the peak temperature during laser welding-brazing is lower, which will help to reduce the mutual diffusion and chemical reaction of Al and Fe atoms, thereby inhibiting the formation of IMCs. Secondly, the melting of steel is avoided during the welding process, which further reduces the possibility of metallurgical reaction. In recent years, scholars at home and abroad have devoted themselves to studying the effects of heat input and metal elements on the IMCs layer and mechanical properties of Al/Fe laser welding-brazing joints.

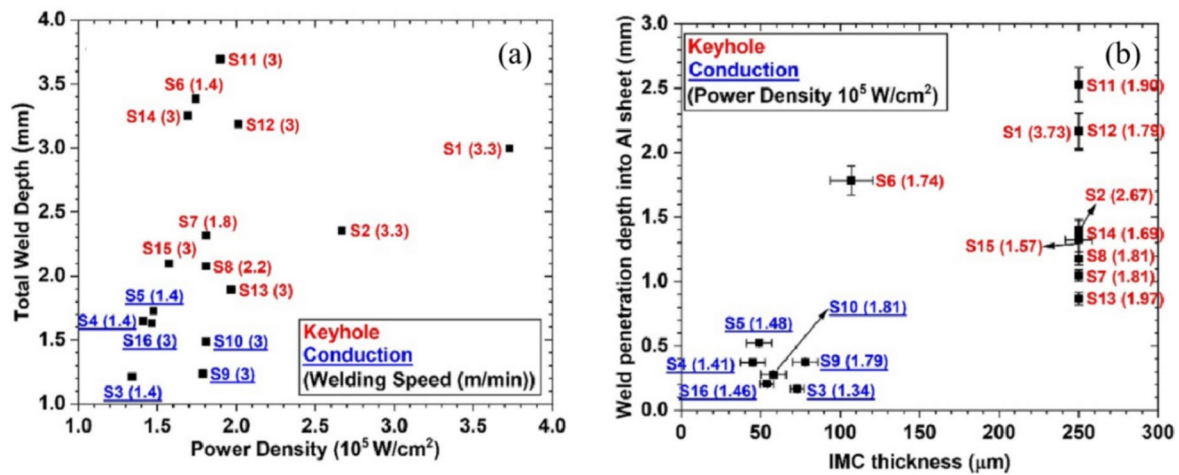


Fig. 6 (a) Shows the effect of energy density on the penetration depth at the speed; (b) the relationship between IMCs thickness and penetration depth under energy density is marked [42]

3.2.1 Control of heat input

Peyre et al. [35] studied the laser welding-brazing joints of DC04 low carbon steel and 6016 Al alloy without the use of brazing filler metal. Unlike the majority of cases where Al is placed below the steel, the authors positioned the aluminum above the steel to achieve a fillet weld. After laser irradiation, the edge of the aluminum base material melted, then wetted and spread onto the steel surface, forming a brazed joint upon solidification. They identified an IMCs layer composed of Fe_2Al_5 with a thickness ranging from 2 to 10 μm and noted that locally thicker IMCs layers were prone to cracking. The authors also investigated the scenario when using galvanized steel: they observed improved wetting of aluminum, which they attributed to the diffusion of molten Al onto the liquid Zn. Springer [56] and Jia et al. [57] also believe that the Zn coating on the steel surface is beneficial during the solid-liquid joining process, which can help the bonding through rapid dissolution, thereby controlling the formation of a joint with uniform chemical composition and no defects. H.R. Kotadia [42] performed laser welding-brazing on IF steel with a Zn coating and 1050 Al alloy, achieving excellent brazed joints at a laser power of 3780 W and a welding speed of 1.4 m/min. The results showed that under a large temperature gradient, near the Al interface, elongated columnar grains formed due to directional heat exchange; in the presence of Al solute, coarse equiaxed grains were formed. Through SEM-EDX and EBSD analysis, three primary IMCs were identified at the interface: $\eta\text{-Al}_3\text{Fe}_2$, $k\text{-AlFe}_3$, and $\theta\text{-Al}_{13}\text{Fe}_4$. Among these, $\eta\text{-Al}_3\text{Fe}_2$ was the main component of the IMCs reaction layer, and these IMCs did not form a continuous reaction layer, as shown in Fig. 7.

Unfortunately, the heat source of single-beam laser brazing is relatively concentrated, resulting in high peak temperatures and a narrow range of adjustable process parameters. Dual-beam lasers, which allow for the separate adjustment of the shape, spot size, energy distribution, and spot distance of the two beams, have been applied in the welding of dissimilar metal materials [36, 58]. Yan et al. [36] used a dual-beam structure consisting of continuous wave (CW) and pulsed wave (PW) lasers to perform laser overlap welding of A6611 aluminum alloy and JSC270CC steel. The principle and schematic diagram of dual-beam laser welding are shown in Fig. 8a and b, respectively. The results indicated that, compared to single-beam welding, dual-beam laser welding can effectively reduce or avoid the formation of porosity in the welded joint. The PW laser beam penetrates the molten pool to form a root-like structure, enhancing the bonding strength at the weld interface, and forming an IMCs layer of 10- μm thickness at the interface, as shown in Fig. 8c and d. Yuan et al. [37] conducted dual-beam laser direct brazing of AA7075 aluminum alloy and DP590 steel. By introducing a derivative laser beam, the wetting angle, wetting distance and thickness of the IMCs layer could be altered. The results showed that with the increase in laser power, the wetting of the aluminum melt on the steel surface improved. When the laser power increased from 1.0 kW to 1.6 kW, the wetting angle decreased from 25.3° to 17.8°, the wetting distance increased from 2.17 mm to 3.19 mm, and the thickness of the IMCs layer increased from 8.37 μm to 12.12 μm .

However, it can be found from the above studies that the IMCs layer at the Fe-Al interface is not completely suppressed by controlling the heat input during the laser welding-brazing process.

Table 4 Effect of alloying elements on the composition and mechanical properties of IMCs in laser welded joints of aluminum and steel

Welding mode	BM	Joint forms	Filler metal	IMCs phase	Mechanical property	Ref
Laser keyhole welding						
	DC56D + ZF steel (1.4 mm)/6016 Al (1.2 mm)	Lap joint	Mn, Zr, and Sn powder	FeAl and FeAl ₃	62.17 MPa max.shear strength	[38]
	08 Al steel (2 mm)/5083 Al (4 mm)	Butt joint	0.1 mm Cu foil	(Fe, Cu) ₄ Al ₁₃ , (Fe, Cu) ₂ Al ₅ and CuAl ₂	2781 N max.tensile strength	[39]
	201 stainless steel (1 mm)/5052 Al (1 mm)	Lap joint	0.1 mm Ni foil	FeAl ₃ and Al _{0.9} Ni _{1.1}	/	[41]
	IF steel (1 mm)/1050 Al (3 mm)	Lap joint	Zn coating	Fe ₂ Al ₅ , Fe ₃ Al and Fe ₄ Al ₁₃	105.3 N/mm ² max tensile strength	[42]
Laser welding-brazing						
	SUS321 stainless steel (3 mm)/5A06 Al (3 mm)	Butt joint	Al-6Cu wire	Al ₁₃ (Fe,Cu) ₄ and Al ₂ Cu	175 MPa max.tensile strength	[44]
	Q235 steel (1 mm)/5052 Al (1 mm)	Lap joint	0.1 mm Cu foil	Fe ₂ Al ₅ , FeAl ₃ and Al ₂ Cu	79.75 N/mm max. tensile strength	[45]
	5 μm Ni-coated Q235B steel (2 mm)/6082 Al (2 mm)	Butt joint	AlSi12 wire	Al ₃ Ni ₂ , (Fe,Ni)Al ₃ and (Fe,Ni) ₂ Al ₃	202.5 MPa max.tensile strength	[46]
	DC04 steel (1.2 mm)/6016-T4 Al (1 mm)	Lap joint	Al-12Si wire	Fe-Al-Si IMCs	190N/mm max. mechanical resistances	[47]
	Galvanized CRRA1000 steel (1.5 mm)/5754 Al (1.5 mm)	Lap joint	Al-12Si wire	Fe(Al,Si) ₃ , Fe ₂ (Al, Si) ₅ and Fe _{1.8} Al _{7.2} Si	345 MPa max. tensile strength	[48]
	DP590 steel (1.2 mm)/6061-T6 Al (1.5 mm)	Butt joint	pure Al, AlSi5 and AlSi12 wire	FeAl ₃ , Fe ₂ Al ₅ , Fe ₂ Al ₈ Si and Fe(Al,Si) ₃	204 MPa max.tensile strength	[49]
	ST07Z galvanized steel (0.8 mm)/5052 Al (1.5 mm)	Lap joint	Al powers	Fe ₂ Al ₅ Zn _{0.4}	/	[50]
	Galvanized DP600 steel (0.77)/6016-T4 Al (1.2 mm)	Lap joint	Zn-15Al wire	Fe-Al-Zn IMCs	227 MPa max.tensile strength	[51]
	DP 980 steel (1 mm)/5754 Al (1.5 mm)	Lap joint	pure Al and Zn-22Al wire	Fe ₂ Al _{5-x} Zn _x and FeZn ₁₀	1230 ± 60 N tensile strength	[52]
	SPCC-GA steel (1 mm)/A5052 Al (1 mm)	Lap joint	Al-Si flux-cored wire and 50 μm Ti foil	/	185N/mm max. tensile strength	[53]
	St-12 steel (1 mm)/6061-T6 Al (3 mm)	Lap joint	Al0.5FeCoCrNi high entropy alloy interlayers	Fe-Al IMCs	5200N max. shear-tensile strength	[54]
	304 stainless steel (1.5 mm)/6061 Al (1.5 mm)	Butt joint	CoZnCuMn0.8Si0.2 powders, FeCoCrNiMn high-entropy alloy powders and Al-12Si powders	Fe-Al IMCs	/	[55]

3.2.2 Addition of filler materials

Based on the principle of alloying, there are typically two methods for adding filler materials in Al/Fe laser welding-brazing: interlayers and filler wires. Adding alloying elements can control the interfacial reaction, mainly reflected

in two aspects: First, it enhances the wetting and spreading of the braze on the surface of the steel; Second, it regulates the growth of IMCs, including their type and thickness, to enhance the properties of the IMCs and reduce their brittleness; it also provides a substrate for heterogeneous nucleation, which aids in refining the grain structure. Currently, the

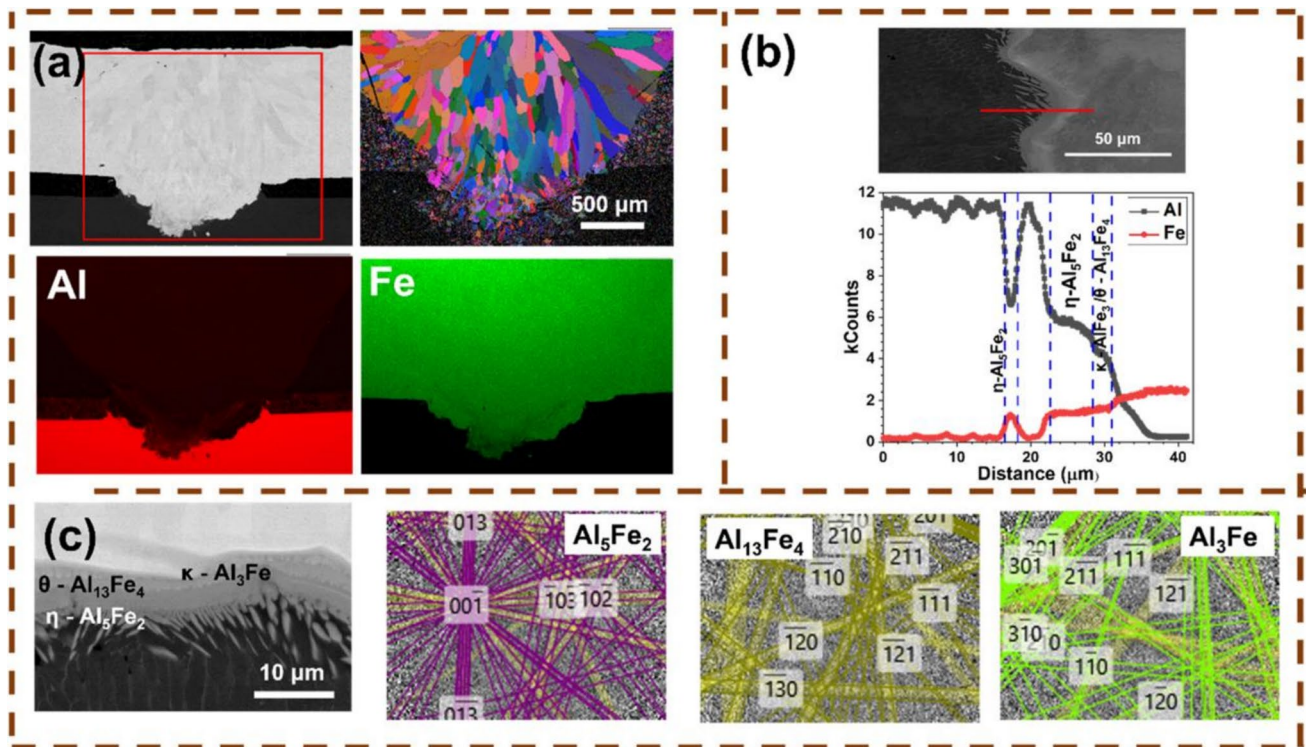


Fig. 7 The microstructure and composition analysis of the joint: (a) EBSD and EDS images show columnar grain growth and IMCs in the weld pool; (b) EDS line scanning results obtained along the red line;

(c) EBSD Kikuchi diagram identified three IMCs phases at the interface between aluminum and steel: η -Al₅Fe₂, θ -Al₁₃Fe₄ and κ -AlFe₃ [42]

alloying elements that have been extensively studied include: Si, Zn, Cu, Cr, Mn, Ni, and high-entropy alloys (HEAs).

The Cu element can exert a certain regulatory effect on the IMCs at the Fe-Al interface. Boag et al. [59] from the Royal Melbourne University of Technology in Australia found that Cu element can be dissolved in Fe-Al compounds to improve the mechanical properties of the compounds. Additionally, the Cu element reacts with the molten Al to form IMCs such as CuAl₂ and CuAl, reducing the brittleness of the IMCs at the interface and improving the mechanical properties of the IMCs [44]. Chen et al. [45] used single/dual-beam lasers, adding a copper interlayer to perform dissimilar metal laser welding-brazing on 5052 Al alloy and Q235 low carbon steel. The authors point out that the microstructure of the joint is composed of a serrated fusion zone with Fe-Al interface and a contact reaction brazing zone with Al-Cu interface. The Fe-Al interface is mainly composed of α -Al and Al₂Cu eutectic, a certain amount of Fe-Al IMCs Fe₂Al₅ and FeAl₃. The Al-Cu interface is mainly composed of Al₂Cu eutectic phase and Al-Cu intermetallic compound metastable phase.

During the welding metallurgical reaction process, the Ni element reacts with Al in the molten pool to form Al-Ni phases, reducing the formation of brittle Fe-Al phases. Other than that, the Ni element refines the grain size, thereby

improving the mechanical properties of the joint [60]. Zhang et al. [46] prepared a Ni coating on the steel surface and used AlSi12 wire, by Cold Metal Transfer (CMT) brazing technology to connect 6082 aluminum alloy and Q235 steel. The study found that the Ni coating improves the wetting and spreading of the AlSi12 solder on the steel surface, giving the Al/Fe joint a good appearance. Compared to joints without a Ni coating, the wetting angles of the joints with 5 μm and 10 μm Ni coatings decreased by 9° and 15.4°, respectively, (Fig. 9a–c). Meanwhile, Ni atoms react with Fe and Al atoms to form Al₃Ni₂, (Fe,Ni)Al₃, and (Fe,Ni)₂Al₃, Fig. 9d presents the XRD analysis results of the joint. When the Ni coating thickness was 5 μm, the tensile strength of the joint reached a maximum value of 202.5 MPa. He et al. [61] applied Ni powder to TIG brazing of A and stainless steel, studied the effect of Ni powders on the joint. He pointed out that at the interface, the effect of Ni powders significantly reduced the thickness of the IMCs layer. At the weld, after adding Ni, the precipitated phase changed from FeAl₆ to Al₉FeNi, and the number of precipitated phases decreased from 12 to 9%. Du [62] from Jilin University utilized a Ni/Cu composite interlayer and achieved laser welding-brazing of TWIP steel and 5083 aluminum alloy by offset the laser beam. In the joints with the added Ni/Cu composite interlayer, a bimetallic solid phase layer of Ni/Cu was formed.

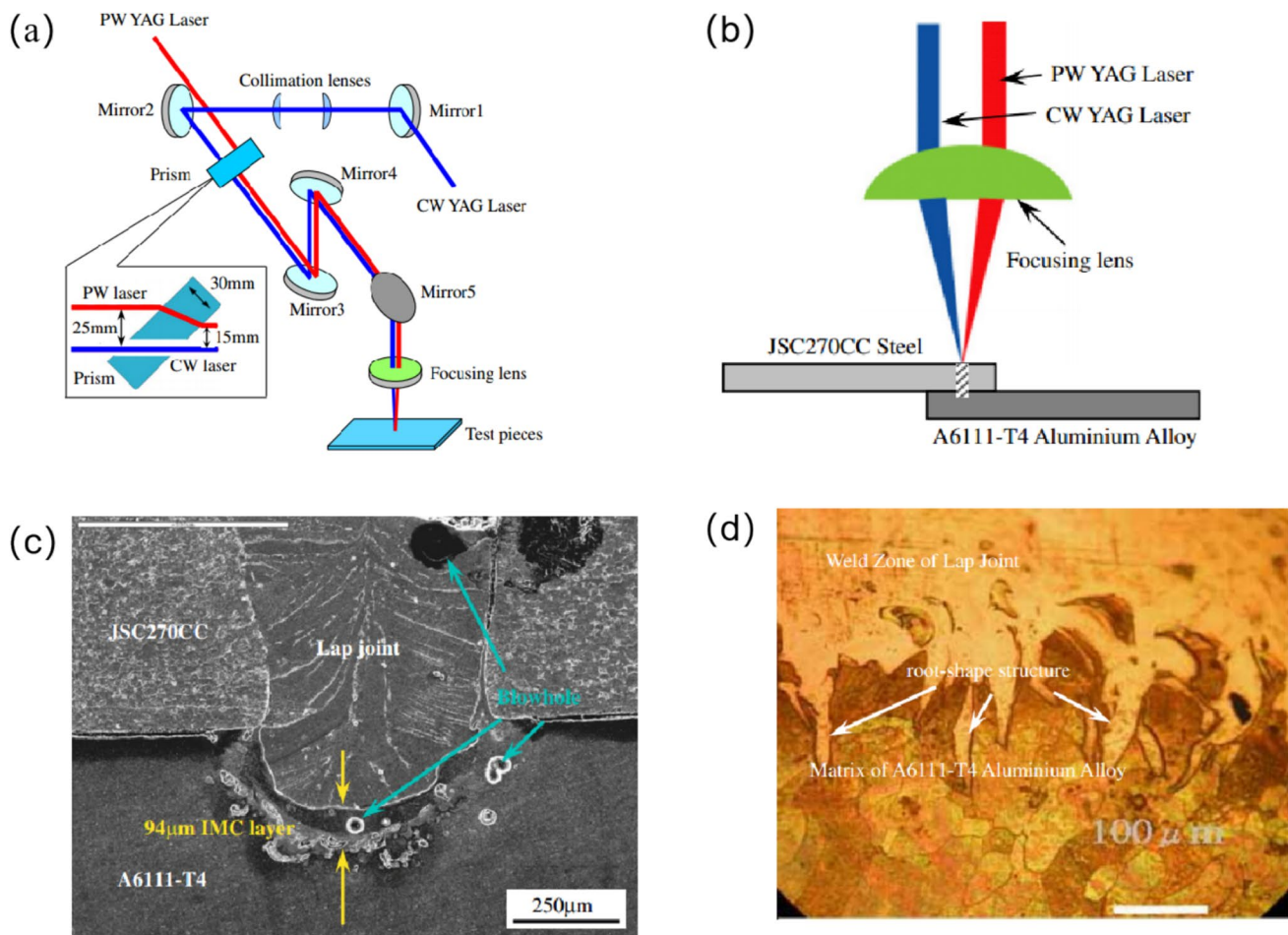


Fig. 8 (a) Principle of dual-beam laser welding; (b) welding schematic diagram; (c) the morphology of single-beam laser welded lap joints; (d) lap joint morphology of dual-beam laser welding [36]

The formation of this bimetallic solid phase layer reduced the content of Fe-Al/Al-Cu IMCs within the weld seam and improved the tensile strength and elongation after fracture of the joint. Compared to joints without the addition of inter-layer materials, the tensile strength of the joints with the Ni/Cu composite interlayer increased by 19.8%.

The addition of Si element has a good inhibitory effect on the formation and growth of IMCs. Besides, it increases the wetting ability of the liquid metal on the steel surface, making it one of the most studied alloying elements by scholars both domestically and internationally. Si atoms tend to occupy the structural vacancies along the c-axis of Fe_2Al_5 , reducing the atomic mobility of the IMCs [63]. This further results in the formation of ternary Fe-Al-Si IMCs and retards the diffusion rate, thereby suppressing the rapid growth of Fe_2Al_5 . Sierra et al. [47] performed laser welding-brazing of 6016-T4 aluminum alloy to DC04 steel using AlSi12 filler wire. The study found that the Fe-Al-Si IMCs layer at the interface had been thinned to less than 2 μm , and the tensile strength of the joint reached 73% of that of the Al alloy BM.

Ding et al. [48] conducted laser welding-brazing of 5754 Al alloy to CRRA1000 ultra-high-strength steel with the aluminum on top and the steel below, using AlSi12 filler wire, and studied the impact of different proportions of the laser spot on the aluminum and steel surfaces. They noted that a larger proportion of the laser spot on the steel surface and higher heat input led to the formation of coarse IMCs enriched with Fe, Al and Si elements near the notch area. By reducing the ratio of the laser spot size to heat input, they successfully formed a uniform IMCs layer at the interface between the weld seam and the steel.

In order to investigate the phase composition of the IMCs layer formed by adding Si element, Yang et al. [64] used diode laser welding technology with AlSi12 filler wire to perform an overlap weld on 5754 aluminum alloy and DP980 steel. The study found that the ternary Fe-Al-Si IMCs formed differed under varying laser powers. Under low laser power (1.4 kW), a layer of IMCs composed of $\theta\text{-Fe}(\text{Al},\text{Si})_3$ and $\tau_5\text{-Al}_{7.2}\text{Fe}_{1.8}\text{Si}$ was observed at the interface between the fusion zone and the steel. At high laser power

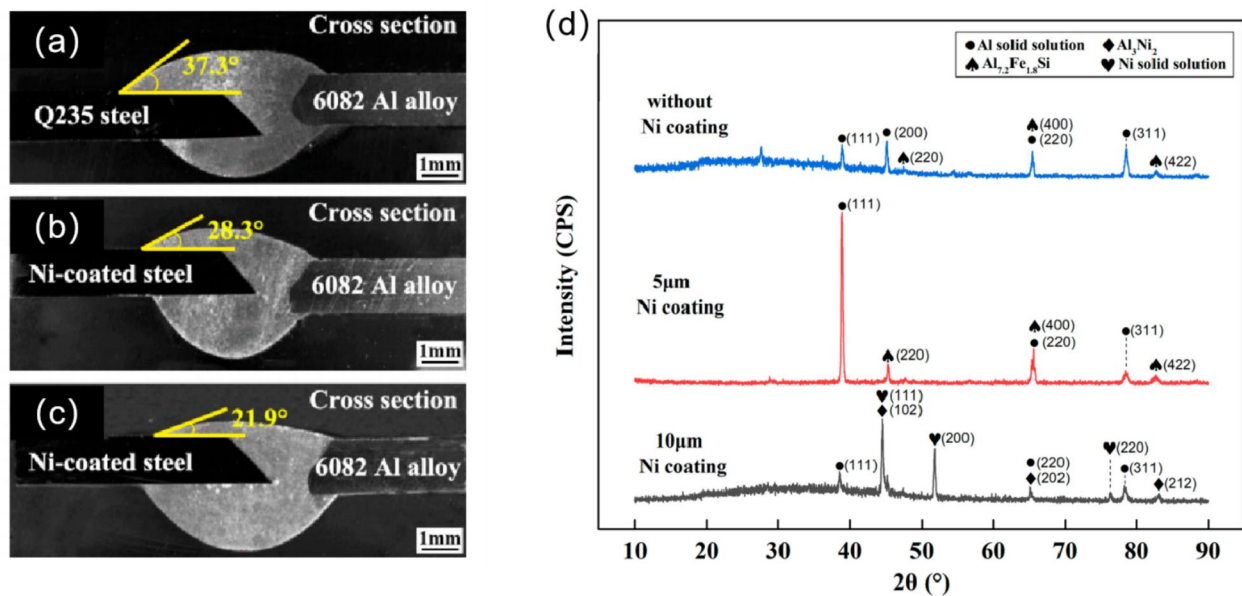
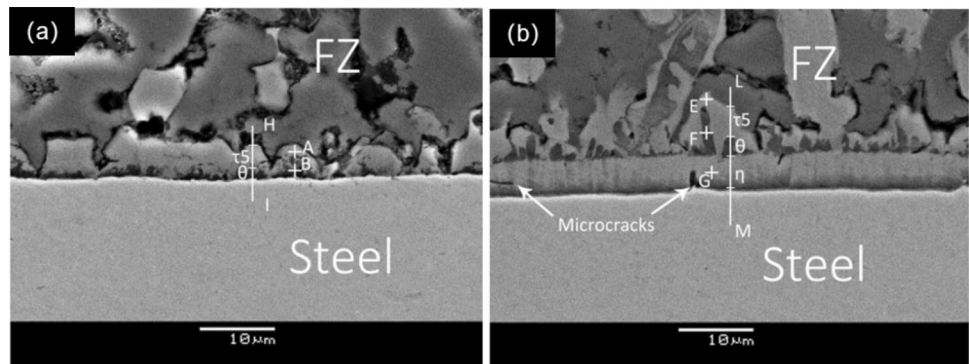


Fig. 9 (a) Macro weld without Ni coating; (b) macroscopic weld of 5-μm Ni coating; (c) macroscopic weld of 10-μm Ni coating; (d) XRD analysis results of welded joints [46]

Fig. 10 (a) SEM image of a welded joint with a laser power of 1.4 kW; (b) SEM image of a welded joint at a laser power of 2.6 kW [64]



(2.6 kW), the microstructure at the interface underwent significant changes. In addition to the original two IMCs, a hard brittle η -Fe₂(Al,Si)₅ phase with microcracks was formed at the interface between the weld seam and the steel. The SEM image of the welded joint is shown in Fig. 10. Xia [49] successfully performed laser welding-brazing of 6061-T6 aluminum alloy without a bevel to DP590 steel with a 45° bevel, using pure Al, Al5Si, and Al12Si filler wires, Fig. 11a. They obtained binary Fe-Al IMCs under conditions with pure Al filler wire and ternary Fe-Al-Si IMCs under conditions with silicon-containing filler wires. The study found that the interface prepared with pure Al filler wire was composed of η -Fe₂(Al,Si)₅ and rod-like θ -Fe(Al,Si)₃, with an average thickness of 10.1 μm at the bottom and 16.7 μm at the top, as shown in Fig. 11c–e. When Al5Si was used as the filler metal, the IMCs composition remained largely unchanged, but the thicknesses at the bottom and top of the

interface decreased to 3.8 μm and 7.5 μm, respectively, as shown in Fig. 11f–h. When Al12Si filler wire was used, the interface IMCs attained a thickness of 1.2 μm of τ_5 -Fe₂Al₈Si at the bottom and a thickness of 5.6 μm of θ -Fe(Al,Si)₃ and τ_5 -Fe₂Al₈Si at the top, as shown in Fig. 11(i–k). Additionally, they conducted tensile strength tests on joints made with different filler metals. The results showed that the joints made with Al5Si filler wire had the highest tensile strength and the greatest displacement at failure (204 MPa and 1.18 mm, respectively). The authors believe that one reason the joints prepared with Al5Si brazing filler have the highest joint strength and the greatest plasticity is due to the appropriate IMCs thickness (less than 10 μm) and the interface IMCs composition of (η -Fe₂(Al,Si)₅) and (θ -Fe(Al,Si)₃) phases, which produce higher interfacial strength; secondly, it is because the IMCs contain Si elements with lower hardness.

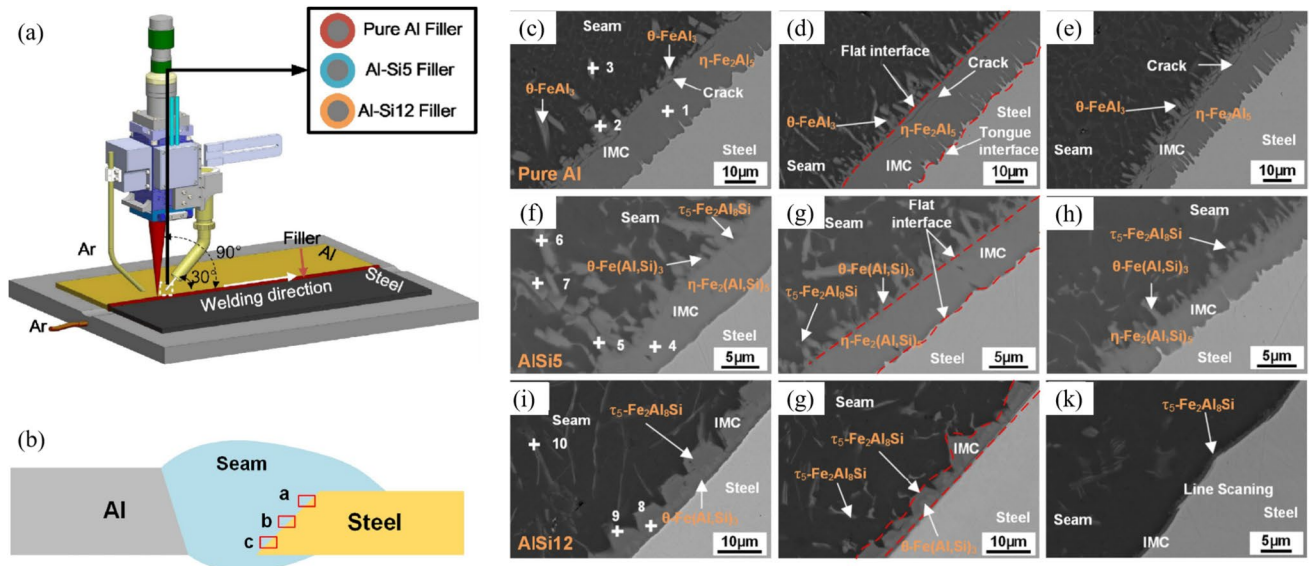
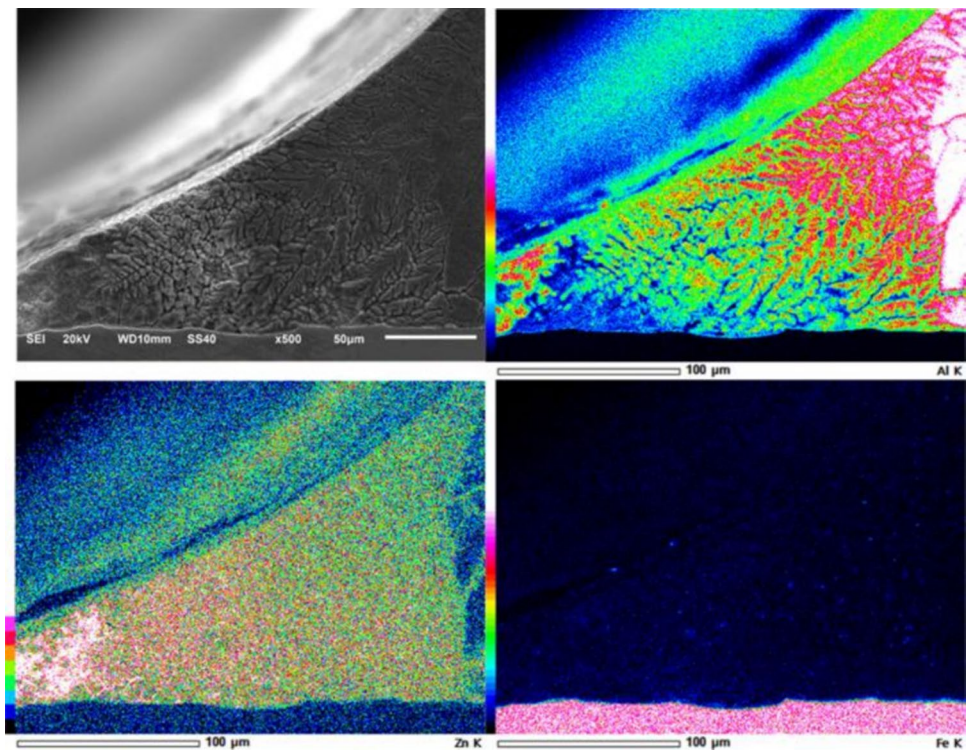


Fig. 11 (a) Schematic diagram of laser fusion brazing of aluminum steel; (b) observation area at the interface between weld and steel; (c–e) microstructure of pure Al wire joint; (f–h) microstructure of AlSi5 wire joint; (i–k) microstructure of AlSi12 wire joint [49]

Since the solubility between Al and Zn is much better than that between Al and Fe, researchers have made a lot of attempts to improve the IMCs at the Al/Fe interface by adding Zn element. Liu et al. [50] performed laser overlap welding of 5052 Al alloy and ST07Z galvanized steel, studied the effect of the Zn layer on the growth behavior of IMCs. The authors noted that the formation process of the $\text{Fe}_2\text{Al}_3\text{Zn}_{0.4}$

phase starts with the formation of Fe_2Al_3 and Al-rich FeAl_3 phases; subsequently, Zn atoms slowly diffuse into the Al–Fe phases, replacing Fe atoms. Figure 12 illustrates the distribution of Zn-rich zones in the joint. Besides, the galvanized layer significantly alters the initial brazing conditions, lowering the brazing temperature from near the melting point of Fe (1535.0°C) to near the melting point of Al (660.4°C).

Fig. 12 The microstructure and basic distribution of zinc-rich zone [50]



Saida et al. [65] also pointed out that a Zn coating can act as a flux to enhance wetting and improve the strength of the joint. Dharmendra [51] used continuous wave Nd:YAG laser brazing technology, adding Al15Zn filler wire to weld galvanized steel (DP600) and aluminum alloy (AA6016). The welded joint exhibited typical characteristics of a brazed and fused double feature: a typical fusion joint formed between the Al alloy side and the brazing filler, while there was no noticeable melting on the steel side. The author noted that the melting time is the primary factor limiting the thickness of the intermetallic reaction layer, and with an increase in brazing speed, the thickness of the reaction layer gradually decreases.

Tan et al. [66] conducted laser butt welding-brazing of AA6061 aluminum alloy with a 45° bevel and DP590 steel without a bevel using flux-cored wires containing Zn-2%Al, Zn-15%Al and Zn-22%Al. The results indicated that the IMCs in the joints obtained using the three types of filler wires were primarily composed of layered η -Fe₂Al₅Zn_{0.4} phases and two different types of δ -FeZn₁₀ phases. Additionally, when using Zn-2%Al and Zn-15%Al, a continuous δ -FeZn₁₀ layer was observed on the steel side, whereas in the joint with Zn-22%Al filler wire, the continuous δ -FeZn₁₀ layer disappeared. The authors explained this phenomenon from a thermodynamic perspective: with the increase in Al content, the Gibbs free energy of the δ -FeZn₁₀ phase increases, and the deficiency of Fe and Zn atoms at the interface is the main reason for the disappearance of the continuous δ -FeZn₁₀ phase near the steel substrate. Due to the disappearance of the continuous δ -FeZn₁₀ phase and the crack-suppressing effect of dispersed δ -FeZn₁₀ phases in the layered Fe₂Al₅Zn_{0.4} matrix, the joint prepared with Zn-22%Al filler wire exhibited the highest tensile strength, reaching 274 MPa. Similarly, Yang et al. [52] found that dispersed δ -FeZn₁₀ phases form within the dominant layered η -Fe₂Al₅Zn_{0.4} phase. The dispersed δ -FeZn₁₀ phases possess lower hardness and better ductility. During tensile loading, the dispersion and softening nature of the δ -FeZn₁₀ phases endow them with higher resistance to cracking, thereby enhancing the strength of the joint. Yan et al. [67] studied the Al-steel laser welding-brazing with added Zn15Al filler wire by external magnetic field. According to EDS analysis, there was no obvious fluctuation in the concentrations of Zn and Al in the interfacial layer. The external magnetic field can effectively control the transport of solutes during solidification. They noted that the enhancement of elemental concentration uniformity by the magnetic field is mainly attributed to the convective effect caused by the thermo-electromagnetic force. During this process, the thermo-electromagnetic force generated by the external magnetic field promotes the flow of the molten pool, thereby eliminating segregation. In addition, the effect of thermo-electromagnetic forces on solid-state diffusion is another cause of enhanced elemental

concentration uniformity by the magnetic field. Comparative analysis shows that the application of an alternating magnetic field during welding can increase the load-bearing capacity of the joint by 12%. In order to control the porosity caused by Zn evaporation in dissimilar joints of 5052 Al alloy and Zn-coated SPCC steel, Ogura et al. [53] proposed inserting a 50- μ m thick Ti interlayer between the dissimilar joints. Although thermal deformation of the Ti interlayer was observed, the porosity was significantly suppressed, resulting in much better mechanical properties compared to joints without the Ti interlayer.

In recent years, scholars have discovered that high-entropy alloys (HEAs), due to their unique high-entropy effect, high-temperature sluggish diffusion effect, and structural stability under multicomponent conditions, tend to form single solid solutions when used as filler materials in dissimilar material welds, thereby suppressing the formation of intermetallic compound phases [55, 68]. At present, researchers have conducted applying HEAs in the field of welding dissimilar materials. Wang et al. [69] used CoCrFeNiMn as an interlayer for laser welding of NiTi shape memory alloy to 304 stainless steel. They studied the effect of the CoCrFeNiMn HEAs interlayer on the microstructure and properties of the NiTi shape memory alloy/304 stainless steel joint. The results indicated that the addition of the HEAs interlayer significantly reduced the formation of Fe-Ti IMCs and promoted the formation of (Fe,Ti) solid solutions. On the other hand, by adding the HEAs interlayer, the average microhardness at the joint interface decreased from 732 to 221 HV. The tensile load of the joint reached 175 N, which is six times that of the joint without the addition of the HEAs interlayer. Hamed Azhari-saray et al. [54] performed resistance spot welding of 6061-T6 Al alloy to low-carbon steel ST-12 using Al0.5FeCoCrNi HEAs of different thicknesses as the interlayer. The results indicated that a solid solution formed on the carbon steel/HEAs side, while complex IMCs developed at the interface on the Al/HEAs side, with the thickness of the HEAs interlayer affecting the morphology of the IMCs. Compared to resistance spot welded joints without HEAs interlayer, using a HEAs as an interlayer improved the tensile strength of the Al/steel resistance spot welded joints. This improvement was attributed to the significantly enhanced bonding strength between the IMCs phase and the Al alloy matrix after the addition of the HEAs interlayer. Liu et al. [55] from East China Jiaotong University prepared welding joints of 6061 Al/304 stainless steel via coaxial powder feeding laser deposition welding, studied the effects of CoZnCuMn0.8Si0.2, FeCoCrNiMn HEAs powders and Al-12Si powders on the microstructure and mechanical properties of the joint interfaces. The experimental results indicated that at the Al side interface, when using CoZnCuMn0.8Si0.2 and FeCoCrNiMn powders as brazing fillers, a high-entropy effect

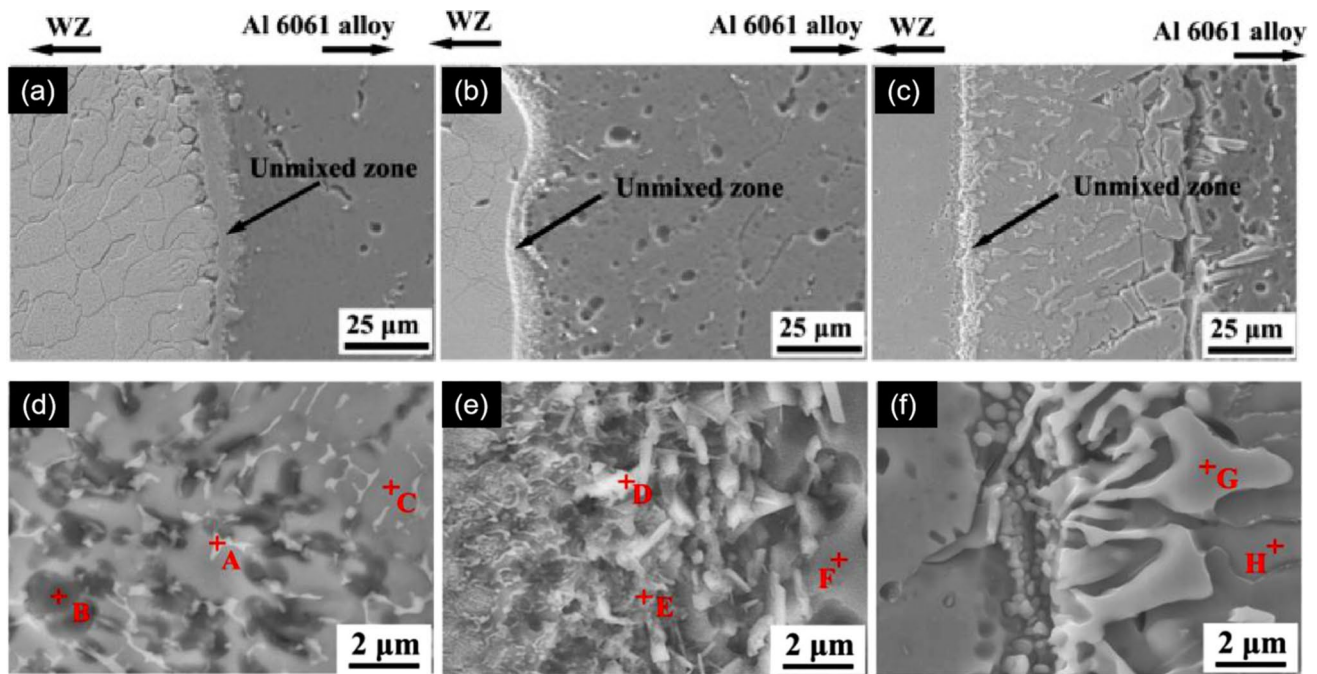


Fig. 13 Microstructure of Al alloy side of different dissimilar joints: (a, d) CoZnCuMn0.8Si0.2 joint; (b, e) FeCoCrNiMn joint; (c, f) Al-12Si joint [55]

occurred at the joint interface, significantly suppressing the formation of Fe-Al IMCs and promoting the generation of fine-grained and equiaxed crystals. In contrast, when using traditional Al-12Si powders as the brazing filler, a large amount of Fe-Al IMCs phases were found at the interface, as shown in Fig. 13a–c. The authors also point out that in traditional Al/steel joints, the IMCs layer is usually needle-like or tongue-like. However, in this study, the Fe-Al IMCs formed by CoZnCuMn0.8Si0.2 and FeCoCrNiMn joints are small in size and dispersed in the unmixed zone, as shown in Fig. 13d–f. The reason may be that the weld zone (WZ) presents a multi-component composition, forming a simple microstructure with high entropy effect, which has obvious hysteresis diffusion effect and reduces the diffusion of Fe element from WZ to Al side.

4 Other laser-related welding technology

In recent years, the development of laser-arc hybrid welding (LAHW) technology has provided a new approach for connecting aluminum alloys and steel with high quality and efficiency. Depending on the type of arc added, LAHW can be categorized into laser-MIG/TIG hybrid welding, laser-plasma arc welding, laser-CMT (Cold Metal Transfer) welding, etc. Compared with single laser welding processes, the primary benefit of using LAHW to connect steel and aluminum alloys is faster speed, deeper penetration, and less

heat input to the surrounding material [70, 71]. Additionally, some scholars have pointed out that LAHW results in reduced formation of IMCs, cracks, thermal distortion, and porosity [72]. However, due to the addition of new forms of heat sources, there are more parameters to control during the welding process, such as current, voltage, wire feed speed, wire-beam distance (the distance between the laser beam diameter and the wire), and arc angle. Therefore, during the LAHW process of dissimilar Al/steel metals, the optimization of welding parameters is particularly important for obtaining high-quality welds.

Qin et al. [71] connected galvanized steel and 6013 Al alloy using large-spot laser-MIG hybrid welding with AlSi5 filler wire. The experimental results showed that at a welding current of 62A, welding speed of 1 m/min, and a laser power of 1 kW, an IMCs layer of 2–4 μm was produced at the brazing interface, with the interfacial phase composition being FeAl_2 , Fe_2Al_5 , and $\text{Fe}_4\text{Al}_{13}$. The maximum tensile strength of the joint reached 247.3 MPa. Meng et al. [73] studied the laser-arc hybrid welding of Al and steel, using beam oscillation to suppress the formation of IMCs and improve the uniformity of the IMCs. Wang et al. [74] applied laser-TIG hybrid welding to aluminum-steel rivet welding, with a schematic diagram of the welding process shown in Fig. 14a. Comparing the performance of joints welded by single-pulse laser and laser-TIG hybrid welding, the distribution of Al and Fe elements in the laser-TIG hybrid welded joint and the single-pulse laser welded joint is shown in Fig. 15. Due

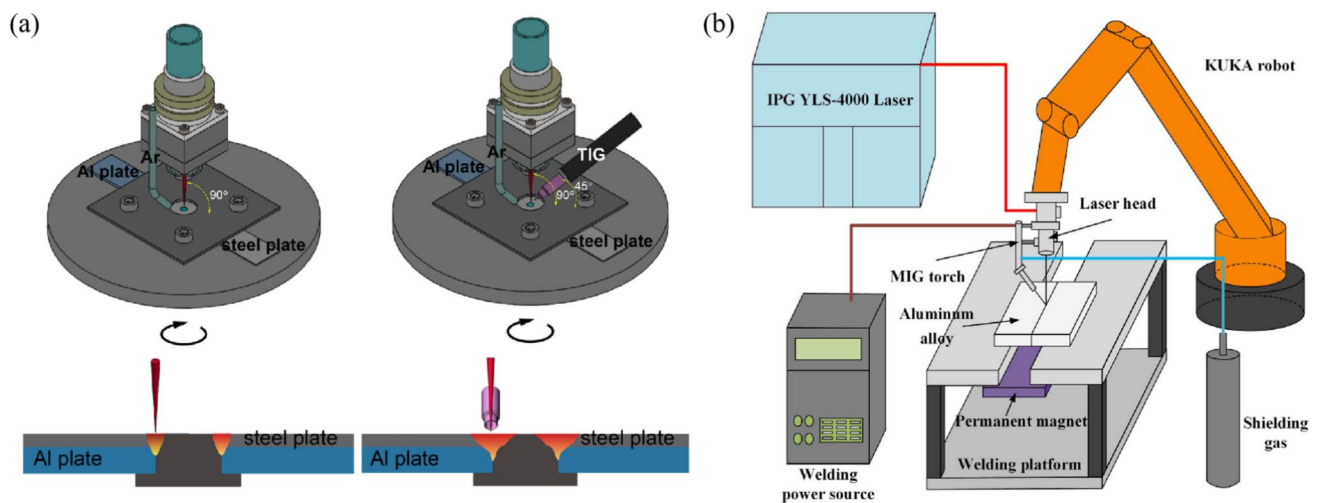


Fig. 14 (a) The schematic diagram of pulsed laser and laser-arc hybrid welding process; (b) external magnetic field assisted laser-MIG hybrid welding experimental device [74, 78]

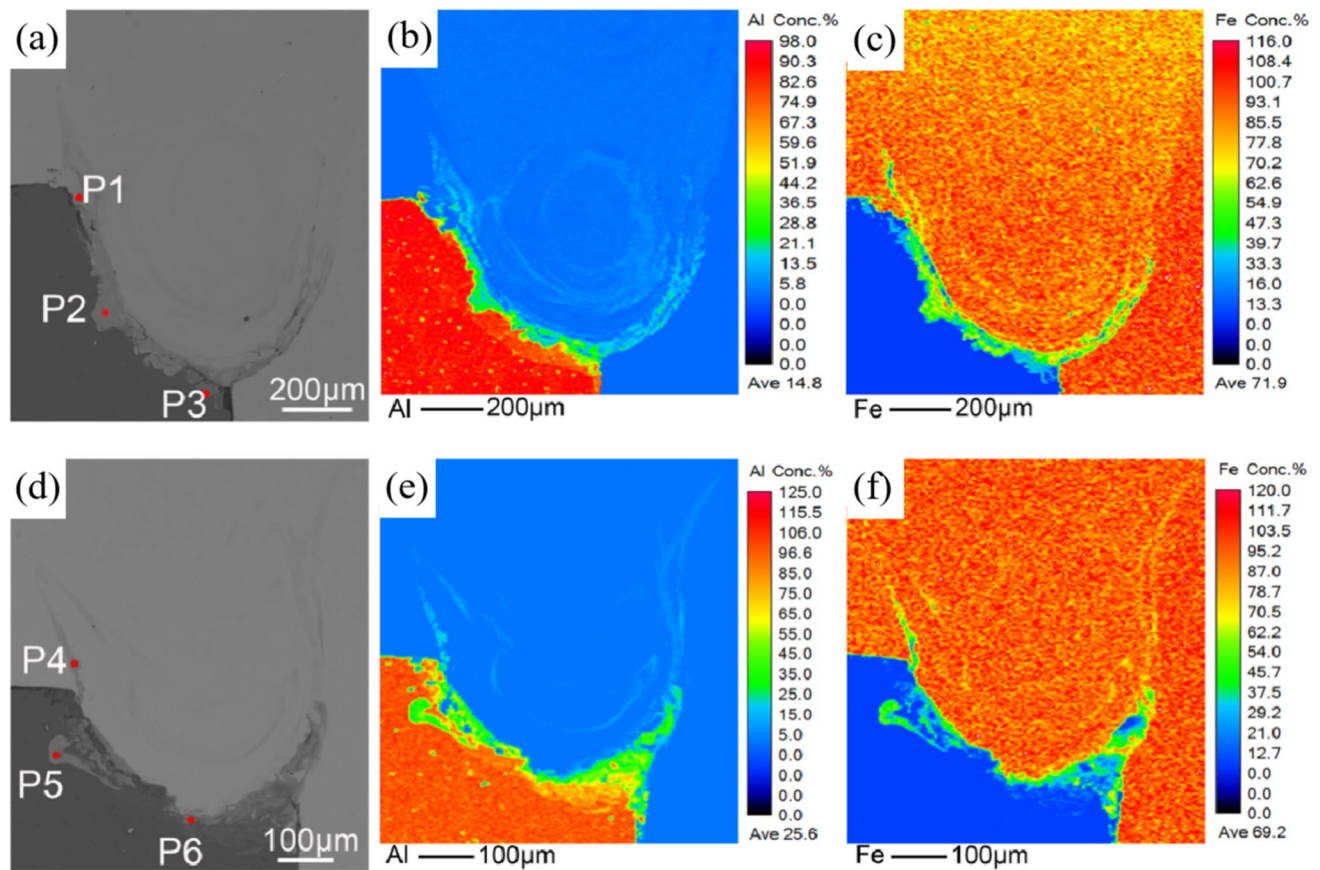


Fig. 15 (a–c) a is the EDS analysis point of pulsed laser welded joint and b, c is the distribution of Al and Fe elements; (d–f) d is the EDS analysis point of LAHW joint and e, f is the distribution of Al and Fe elements [74]

to the high pulse energy and pulse overlap of the pulse laser, the molten pool experiences intense stirring by the thermal effect. Under the action of convection, Al elements continuously diffuse towards the steel fusion zone, with some Al elements gathering in a vortex-like manner at the lower part of the weld. The stirring effect of the molten pool by LAHW is weaker than that of the pulse laser with high pulse energy, thus exerting a lesser driving force for the diffusion of Al elements. Al elements mainly concentrate at the bottom of the fusion zone, with small amounts of Al distributed in chunks or fine strands within the Al-steel mixed zone. Compared to standalone pulsed laser welding, the heat source in LAHW achieves a larger molten pool on the steel plate, reducing the area of Al melting while ensuring the complete melting of high-strength steel. It not only effectively reduces the formation of IMCs, but also makes the distribution of IMCs more dispersed, thereby improving the overall performance of the Al/steel riveted hybrid joint structure. Besides, external magnetic fields have also been utilized in LAHW, as shown in Fig. 14b. By introducing a magnetic field, the Lorentz force is generated in the molten pool, which stirs the molten pool, homogenizes the composition of the weld zone, and improves the quality of the joint. Other than that, it reduces the probability of keyhole collapse and inhibits pore formation. Secondly, due to the driving effect of the magnetic field on the liquid metal, Fe-Al compounds are distributed diffusively at the interface [75–78].

Due to the extremely low thermal input of the CMT arc into the BM, laser-CMT weldig can reduce the negative impact of adding an arc heat source on the interfacial reactions compared to laser-MIG hybrid welding [79]. In addition, the hybrid heat source achieves high welding speeds. Because of the high heating and cooling rates, interfacial reactions can be suppressed. Chen et al. [80] used laser-CMT technology to perform butt welding of 5052 Al and Q235 steel. The study found that the wire feeding speed affects the formation of pores, the shape of the joint, and the thickness of the IMCs layer; as the wire feeding speed increases, the thickness of the interfacial IMCs layer also increases. The offset distance has a significant impact on the joint strength: when the offset distance is 0.2 mm, the thickness of the IMCs layer is too high, leading to spontaneous cracking of the joint; on the other hand, an excessively large offset distance results in poor wetting of the Al melt on the steel surface. Additionally, a similar effect was observed regarding welding speed: the faster the welding speed, the thinner the IMCs layer due to reduced heat input; however, if the welding speed is too high, the wetting is impaired. The IMCs layer consists of Fe_2Al_5 closer to the steel side and needle-like $\text{Fe}_4\text{Al}_{13}$ near the weld seam, with a thickness ranging from 3–5 μm .

Laser-assisted friction stir welding (LAFSW) is an improvement on the friction stir welding (FSW) process,

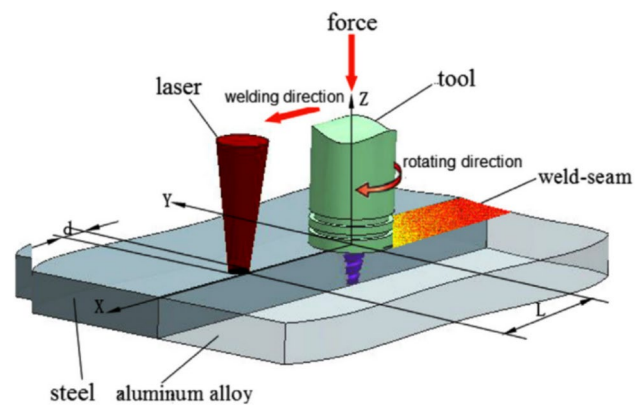


Fig. 16 Laser-assisted FSW process schematic diagram [83]

combining FSW with laser welding. In LAFSW, the workpiece is preheated by adding a high-intensity laser beam ahead of the rotating tool, which can significantly reduce the strength and hardness of the materials to be welded, decreasing the wear of the stirring tool [81–83]. Due to the reduction in temperature, the formation of IMCs is restricted. A typical setup for LAFSW is illustrated in Fig. 16. In recent years, LAFSW processes have primarily been applied to the joining of Al, Ti, Mg, Al/steel and Al/Mg materials.

Chang et al. [84] investigated the microstructure and mechanical properties of LAFSW of 6061-T6 Al to AZ31 Mg alloy. The authors used a Ni foil as filler material to join the Al to the Mg. They pointed out that in the composite welding with the addition of Ni foil, the presence of a minor quantity of Ni-based IMCs layer replaced $\text{Al}_{12}\text{Mg}_{17}$, thereby reducing the amount of brittle IMCs and increasing the joint strength. It was found that when the laser power reached 2 kW, the joint strength was 66% of Mg base material. Merklein et al. [85] performed LAFSW of a 1 mm steel plate and an Al plate. The results indicated that, compared with conventional FSW, this method improved the tensile strength, reduced tool wear, and formed a good joint. Fei et al. [86] used numerical simulation to calculate the preheating temperature field generated by the laser beam on the cross-section of the workpiece during LAFSW of Al and steel, testing the tensile strength of the joints under different laser powers. The results indicated that only with appropriate laser power could the steel achieve the suitable preheating temperature, thereby forming a thinner IMCs layer and giving the joint higher tensile strength; both excessive and insufficient laser power would decrease the tensile strength of the joint. The keyhole offset distance, IMCs layer thickness and type are critical factors affecting the tensile strength of the steel/Al joint in LAFSW. In another study, he pointed out that the keyhole offset distance has a significant influence on the thickness and type of the IMCs layer. An appropriate

keyhole offset distance can form a thin IMCs layer and Fe-rich IMCs, achieving a joint with high tensile strength [83].

5 Application of numerical simulation in laser welding of Al/steel

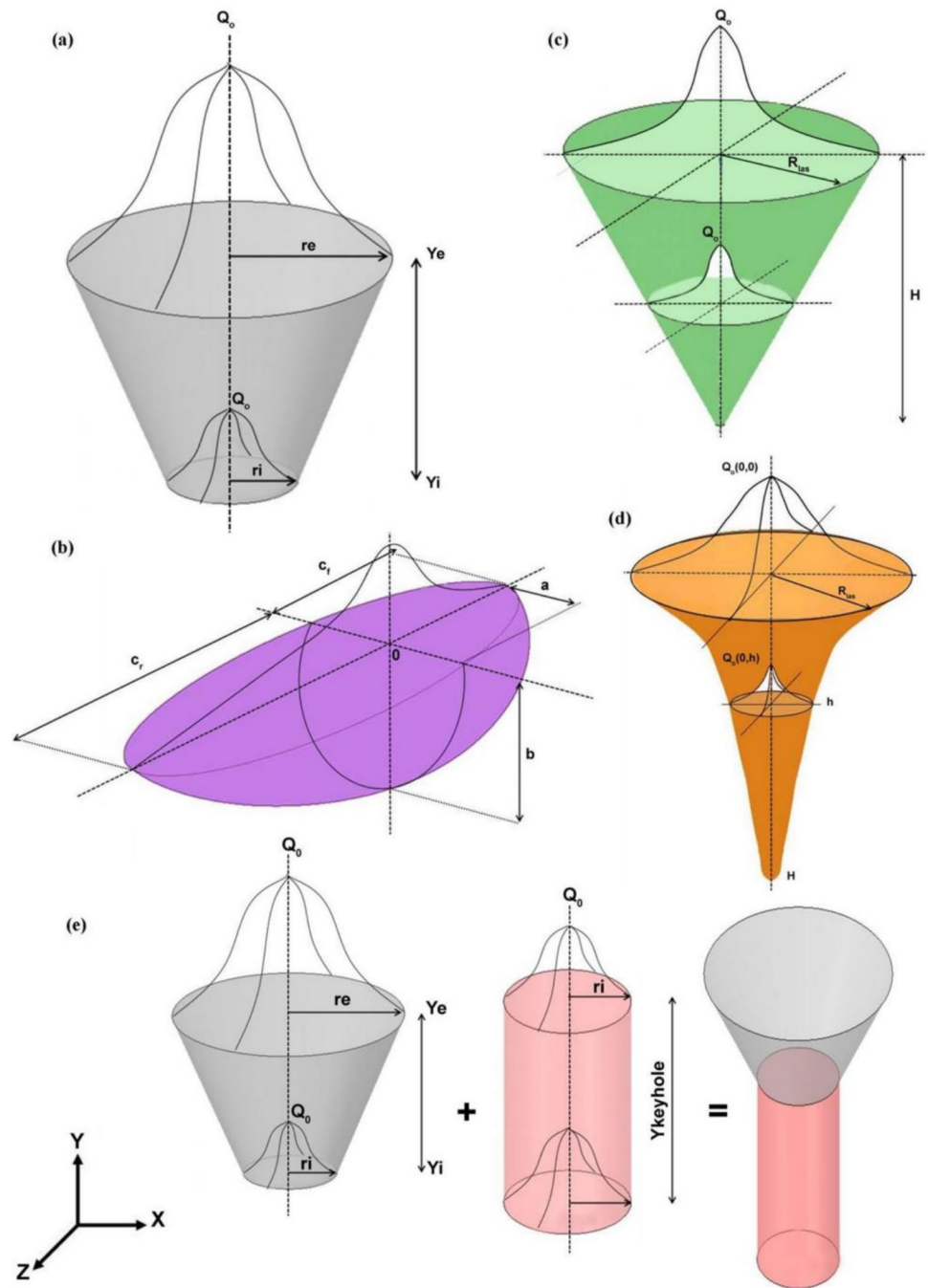
With the development of the computer field, numerical simulation has increasingly found application in the realm of welding. During the laser welding process, the thermal cycles applied to the workpiece lead to non-uniform thermal expansion and contraction, subsequently resulting in residual stresses and deformations. Moreover, welding occurs over an extremely short period, making it exceptionally challenging to study issues such as fluid flow and pore formation at the welding site. Therefore, numerical simulation holds significant importance in the laser welding industry, playing a vital role in the theoretical and process research of laser welding. It aids in understanding the welding mechanism, effectively shortens experimental time, reduces research costs, obtains ideal process parameters, and enhances the quality of the welded joint [87–89]. Following this, the text enumerates the progress in research employing numerical simulation methods to study the temperature field, stress field, and melt flow field in laser welding of Al/steel.

The application of numerical simulation to study welding temperature fields generally includes the following steps: (1) Determining the type of heat source model; (2) Establishing the heat source parameters and boundary conditions; (3) Conducting numerical simulation; (4) Experimental comparison and validation [90]. In the numerical simulation of laser welding, commonly used heat source models include the double ellipsoidal heat source model, Gaussian surface heat source model, Gaussian conical heat source model and combined heat source model. Schematics illustrating different heat source models are shown in Fig. 17 [91–93]. Qin [94] studied the temperature distribution in different regions of the joint during MIG welding of thin 6013 Al sheet to galvanized steel sheet by employing a combined heat source model featuring a double ellipsoidal heat source and a uniform volumetric heat source. In the modeling process, the MIG arc was treated as a double elliptical Gaussian plane heat source, while the molten droplets were considered as a uniform volumetric heat source, taking into account the effects of the Zn coating and the overlapping gap. The computational results indicate that the temperature distribution within the fusion-brazed joint is uneven, and the temperatures on the steel/Al sides are not symmetrically distributed. Compared to the steel side, the Al side exhibits a smaller temperature gradient and a wider high-temperature zone, showing good agreement with experimental results. Cui et al. [95] used ABAQUS software to analyze the thermodynamics of swing laser welding of Al–Mg alloy. The model

grid is shown in Fig. 18a–d. In the process of numerical simulation, he chose the Gaussian cylindrical heat source model and the periodic oscillation mode of the laser beam. The simulation results show that the central area of the heat source moves left and right with the swing of the laser beam, which makes the temperature distribution in the joint more uniform. During the swing of the laser beam, the isotherm density in the central area of the heat source is larger and the temperature gradient is larger. The area away from the center of the heat source is the opposite, Fig. 18e–g. Figure 18h shows the comparison results of the experiment and simulation. The actual melting half-width of the weld is 1.51 mm, and the melting half-width of the weld obtained by numerical simulation is 1.62 mm, with an error of 7.3%. The actual weld melting depth is 2.52 mm, and the weld melting depth obtained by numerical simulation is 2.73 mm, with an error of 8.3%. Additionally, Tan et al. [66] simulated the temperature curves at different positions on the interface of laser-welded Al/steel joints using a Gaussian volume heat source model. Based on this, they evaluated the chemical potential of Zn elements using the Toop model [96] and analyzed the diffusion behavior of Zn in the Fe–Al–Zn ternary system from a thermodynamic perspective.

From Table 1, it can be seen that there is a significant difference in the thermal conductivity and coefficient of thermal expansion between Al and Fe. During the laser welding process, heat concentrates in a very small area, causing rapid temperature rises and localized thermal expansion. In the cooling process, due to the differences in the coefficients of thermal expansion, considerable residual stresses develop in the Al/steel joint. Residual stresses from welded can lead to deformation and crack formation, reducing joint strength and shortening service life [87]. To evaluate the mechanical properties of the joint or to provide a basis for optimizing welding process parameters, the thermodynamics model is established using numerical simulation methods. Based on the obtained temperature field, residual stress fields and strains are calculated using material constitutive models. This allows prediction of the distribution and magnitude of residual stresses and strains before laser welding, thus providing a basis for welding. During the simulation of the welding stress field, the stress–strain model is established based on the theory of thermoelastic–plasticity [97]. Myhr [98] pointed out that for non-ferrous materials, such as aluminum alloys, their mechanical properties and microstructural changes are governed by precipitation kinetics, proposing the use of a constant dissolution rate model to describe this. Sonne et al. [99] analyzed the evolution of residual stress after FSW of 2024 Al. The authors noted that, when predicting residual stresses, considering the softening ratio is more critical than selecting a plasticity model. So far, numerous studies have demonstrated a correlation between the mechanical properties of Al/steel joint and the

Fig. 17 Laser welding heat source model: (a) double ellipsoid heat source model; (b) Gaussian cone heat source model; (c) Gaussian volume heat source model; (d) Gaussian rotating heat source model; (e) Conical-cylindrical composite heat source model [93]



Al concentration within the weld, showing a non-linear increase in weld hardness and yield strength as Al concentration increases [11, 100–102]. Anton Evdokimov [103] considered the impact of Al concentration on the thermodynamics properties of Al/steel joints, combining nonlinear finite element (FE) with temperature field simulations. The material model established effectively predicted the welding deformation of the joint.

In laser welding, the simulation of the keyhole and molten pool flow field can well reveal the intrinsic

mechanisms of weld formation and porosity formation. With the deepening of the keyhole, there can be fluctuations in the effective absorption rate of the laser, and even with a stable keyhole formed, the shape of the keyhole continues to fluctuate. Dual-beam laser welding can significantly reduce the amplitude of keyhole depth fluctuations [104, 105]. The principle of dual-beam laser welding and the transient keyhole model are shown in Fig. 19a–c. Wang et al. [106] proposed a three-dimensional transient model based on arc-Al melt-solid steel

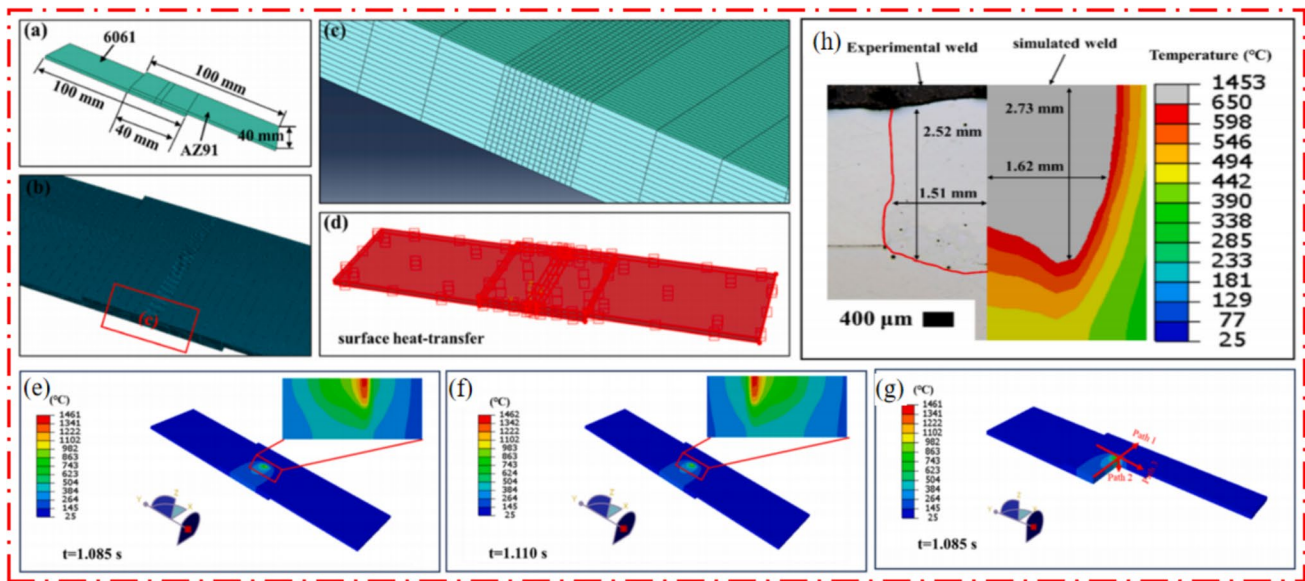


Fig. 18 Simulation of laser welding of Al-Mg alloy: (a–d) grid model; (e–g) the temperature distribution characteristics of the joint during laser oscillation; (h) comparison of test and simulation results [95]

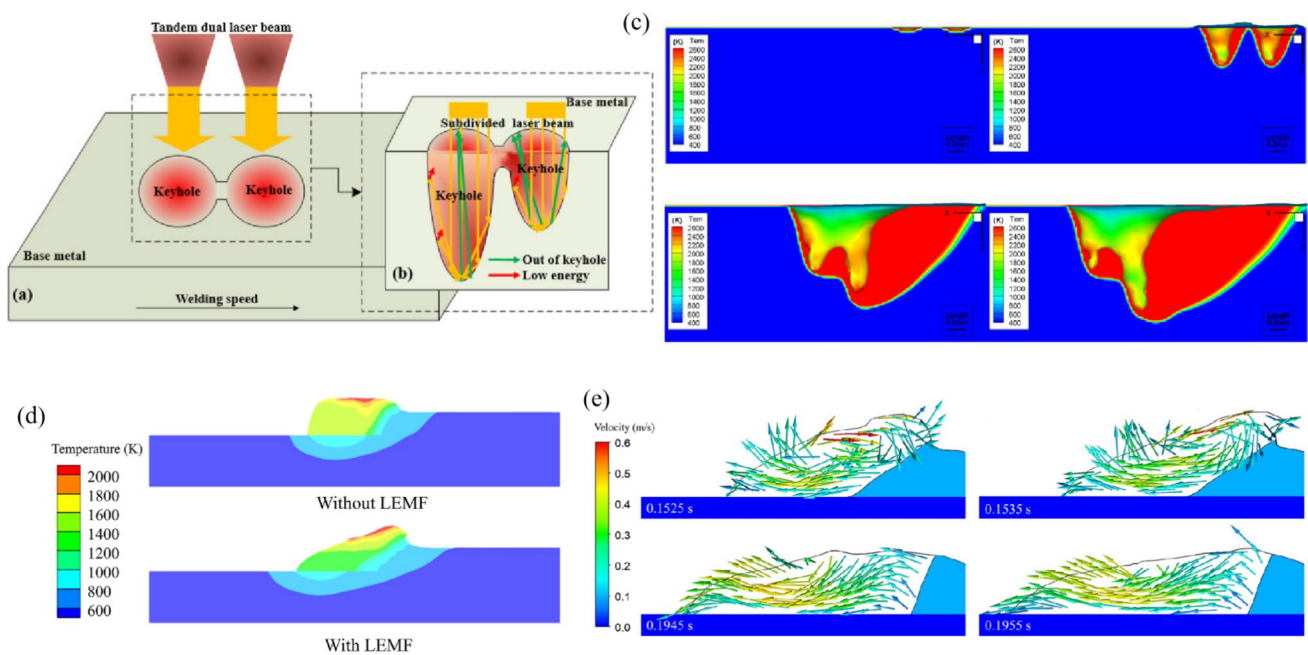


Fig. 19 (a, b) Principle of dual-beam laser welding; (c) two-beam transient keyhole model; (d) temperature curves of molten pool cross section without MF and with MF; (e) fluid flow velocity in molten pool with MF [105, 106]

interactions, taking into account the effects of multiple forces (gravity, electromagnetic, buoyancy, surface tension and Marangoni forces) on fluid flow within the molten pool. This model revealed the mechanism of magnetic field (MF) influence on molten pool behavior during MIG brazing of Al/steel metals, as illustrated in Fig. 19d,e. Analysis indicated that after applying a MF,

the highest temperature on the surface of the molten pool decreased while the width of the molten pool increased. The periodically varying flow pattern further augmented the width of the molten pool and promoted the spread of liquid metal over the steel surface. Unlike magnetic field-assisted laser welding, in magnetic field-assisted LAHW, the Lorentz force is generated by the interaction of the

welding current with the MF. Tan et al. [107] simulated the keyhole dynamics and weld pool behavior of LAHW of aluminum alloy, and found that the stable MF inhibited the deformation of the back wall of the keyhole by droplets and eddy currents, widened the keyhole size, reduced the collapse of the keyhole, and improved the stability of the keyhole.

The grain morphology and microstructural transformations in the weld zone have a significant impact on the mechanical properties of the joint. Combining macroscopic and microscopic simulations for multiscale analysis will greatly advance the study of mechanisms behind the formation of laser-welded seams in Al/steel joint and the prediction of weld defects. Currently, researchers primarily employ Monte Carlo (MC), Phase Field (PF), and Cellular Automata (CA) methods for multiscale analysis. Zhang et al. [108] calculated the formation and growth of Fe-Al IMCs by MC, and revealed the formation and growth mechanism of Al-Fe IMCs at the Al-Fe interface. With the increase of simulation time, the thickness of Fe_2Al_5 IMCs layer increased nonlinearly. Tan and Shin [109] utilized a macroscopic dynamics model to predict fluid flow and heat transfer within the keyhole and molten pool, feeding the generated thermal history data into a CA model to predict grain growth and dendrite morphology during the laser welding of 304 stainless steel. Geng et al. [110] developed a multiscale model combining macro-scale heat transfer and fluid flow modeling with micro-scale PF modeling to investigate the evolution of grain and subgrain structures during the laser welding of 5083 Al, elucidating the development of solidification morphologies during the transition from cellular to dendritic structures, as shown in Fig. 20.

6 Summary and outlook

Laser welding of Al/steel dissimilar metals is one of the important techniques for achieving lightweight design in automobiles. However, due to the significant differences in the physical and chemical properties between Al and steel, brittle IMCs tend to form during welding, which can negatively impact the mechanical properties of the welded joint.

To tackle this challenge, researchers have adjusted welding process parameters to reduce the thermal input during welding, reduced the interface reaction time to adjust the thickness, composition and distribution characteristics of Al/Fe IMCs. From the perspective of alloying, adding filler materials to participate in the metallurgical reactions within the molten pool can effectively suppress the formation of Fe-Al phases. Extensive research cited in the text indicates that both approaches have, to some extent, inhibited the formation of Al/Fe IMCs, thereby enhancing the strength and toughness of the welded joints. However, it is currently not possible to completely eliminate the generation of IMCs during the laser welding of Al and steel.

Future research directions should focus on developing more efficient strategies for controlling IMCs, including but not limited to optimizing welding process parameters, identifying new interlayer materials, and advancing numerical simulation technologies to better understand the thermodynamic and fluid dynamic behaviors during welding. The following approaches are specifically proposed:

- (1) Explore the use of high-entropy alloys (HEAs) as interlayers to suppress the formation of brittle Fe-Al intermetallic compounds (IMCs). This is achieved through the synergistic effects of multiple elements and exten-

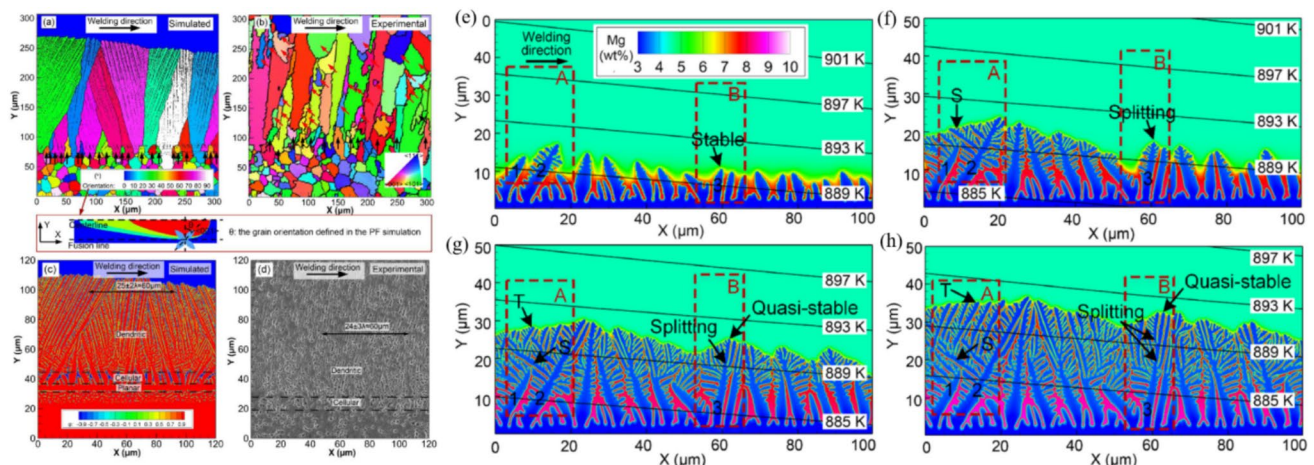


Fig. 20 (a–d) Comparison of grain/subgrain structure between simulation results and experimental results; (e–h) the evolution process of cell to dendrite [110]

sive experimental screening to identify the optimal alloy composition.

- (2) Leverage interdisciplinary research to develop real-time feedback control systems integrated with machine learning algorithms. These systems dynamically adjust welding parameters, including welding speed, laser power, and spot offset, to minimize the thickness of IMCs and enhance joint strength.
- (3) Utilize molecular dynamics (MD) simulations to study atomic diffusion behavior at the Fe/Al interface. Design gradient nanostructured interfaces (e.g., via gradient composition or grain boundary engineering) to reduce stress concentration.
- (4) Investigate the combination of ultrafast lasers (femto-second/picosecond level) with arc welding to achieve precise control of IMC layers. This is done by reducing heat input through ultra-short pulses, while taking advantage of the deep penetration capabilities of arc welding.
- (5) The application of multi-scale analysis methods, including the integration of macroscopic and microscopic simulations, will deepen the understanding of welded joint formation mechanisms and help predict defects.
- (6) Further research on the long-term performance of Al/steel welded joints under real-world conditions is essential to ensure the safety and reliability of welded structures.

Author contribution Qiubo Li.: Conceptualization, Methodology, Writing original draft. Shanglei Yang.: Validation, Visualization, Supervision. Bangguo Hu. and Suqi Xue.: Supervision, Writing—review and editing. All authors have read and agreed to the published version of the manuscript.

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Declarations

Competing interests The authors declare no competing interests.

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